UNIVERSITY of HOUSTON | ECE

ECE 3337: Signals & Systems Analysis

Lecture 23: Fourier Analysis Techniques

Reference: Chapter 5
(Section 5-11 - 5-12)

Please turn off your gadgets now

Recap: The Fourier Transform is valuable for Representing non-periodic Waveforms

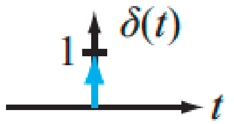
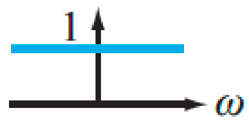
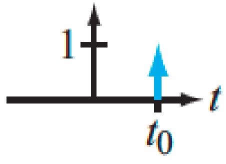
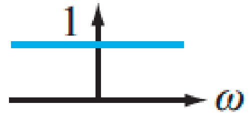
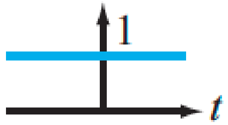
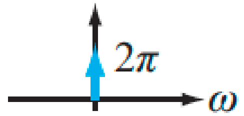
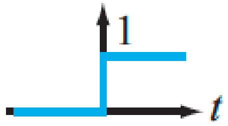
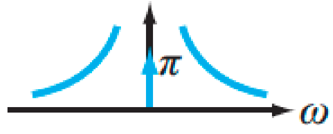
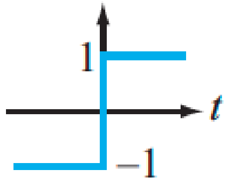
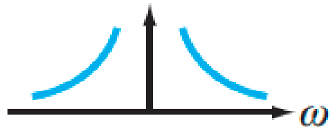
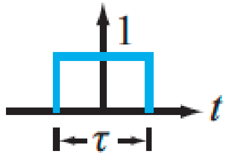
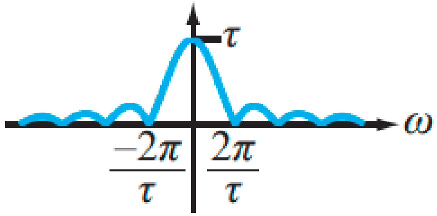
$$\hat{\mathbf{X}}(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

$$x(t) = \mathcal{F}^{-1}[\hat{\mathbf{X}}(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{\mathbf{X}}(\omega)e^{j\omega t}\,d\omega$$

$$x(t) \leftrightarrow \hat{\mathbf{X}}(\omega)$$

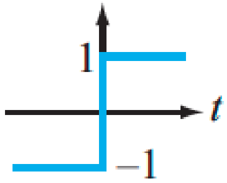
The hat symbol ^ over X distinguishes this from the Laplace Transform

Common Fourier Transform Pairs

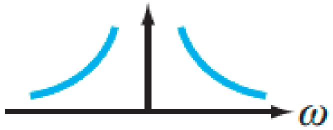
	$x(t)$	$\mathbf{X}(\omega) = \mathcal{F}[x(t)]$	$\lvert\mathbf{X}(\omega)\rvert$
		BASIC FUNCTIONS	
1.		$\delta(t) \leftrightarrow 1$	
1a.		$\delta(t - t_0) \leftrightarrow e^{-j\omega t_0}$	
2.		$1 \leftrightarrow 2\pi\ \delta(\omega)$	
3.		$u(t) \leftrightarrow \pi\ \delta(\omega) + 1/j\omega$	
4.		$\mathrm{sgn}(t) \leftrightarrow 2/j\omega$	
5.		$\mathrm{rect}(t/\tau) \leftrightarrow \tau\ \mathrm{sinc}(\omega\tau/2)$	

Common Fourier Transform Pairs

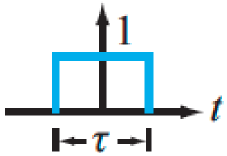
4.



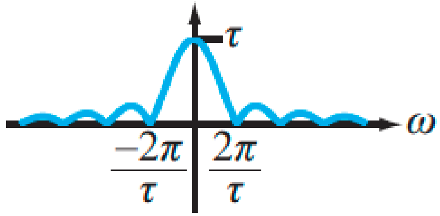
$$\mathrm{sgn}(t) \longleftrightarrow 2/j\omega$$



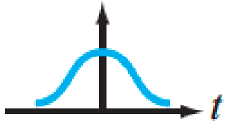
5.



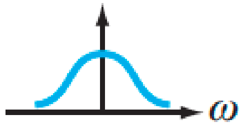
$$\mathrm{rect}(t/\tau) \longleftrightarrow \tau\,\mathrm{sinc}(\omega\tau/2)$$



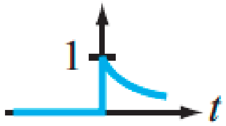
6.



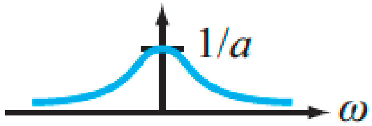
$$\frac{e^{-t^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}} \longleftrightarrow e^{-\omega^2\sigma^2/2}$$



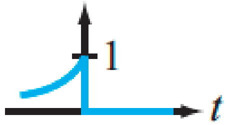
7a.



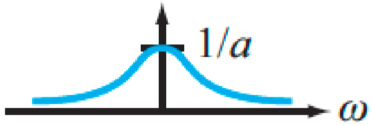
$$e^{-at}\,u(t) \longleftrightarrow 1/(a+j\omega)$$



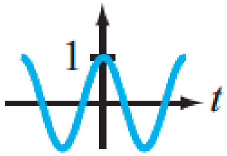
7b.



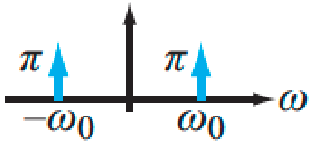
$$e^{at}\,u(-t) \longleftrightarrow 1/(a-j\omega)$$



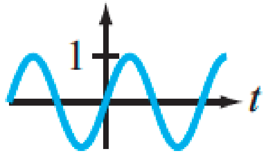
8.



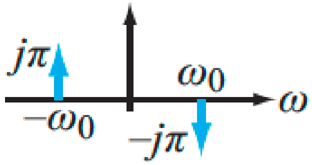
$$\cos\omega_0 t \longleftrightarrow \pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$$



9.



$$\sin\omega_0 t \longleftrightarrow j\pi[\delta(\omega+\omega_0)-\delta(\omega-\omega_0)]$$

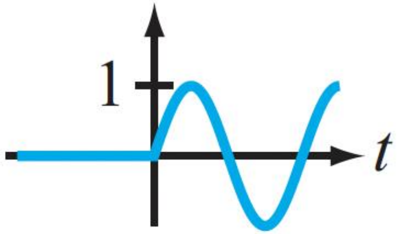


Causal Signals: Fourier is Messier

$$e^{j\omega_0 t} \quad \longleftrightarrow \quad 2\pi\ \delta(\omega - \omega_0)$$

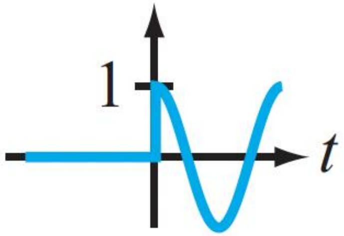
$$t e^{-at}\ u(t) \quad \longleftrightarrow \quad 1/(a + j\omega)^2$$

$$[e^{-at} \sin \omega_0 t]\ u(t) \quad \longleftrightarrow \quad \omega_0/[(a + j\omega)^2 + \omega_0^2]$$



$$[\sin \omega_0 t]\ u(t) \quad \longleftrightarrow \quad (\pi/2j)[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] + [\omega_0^2/(\omega_0^2 - \omega^2)]$$

$$[e^{-at} \cos \omega_0 t]\ u(t) \quad \longleftrightarrow \quad (a + j\omega)/[(a + j\omega)^2 + \omega_0^2]$$



$$[\cos \omega_0 t]\ u(t) \quad \longleftrightarrow \quad (\pi/2)[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + [j\omega/(\omega_0^2 - \omega^2)]$$

MESSY ! (WHERE IS THE MESSINESS ORIGINATING FROM?)

THE UNILATERAL LAPLACE TRANSFORM IS MORE CONVENIENT FOR CAUSAL SIGNALS

Fourier Transform Properties

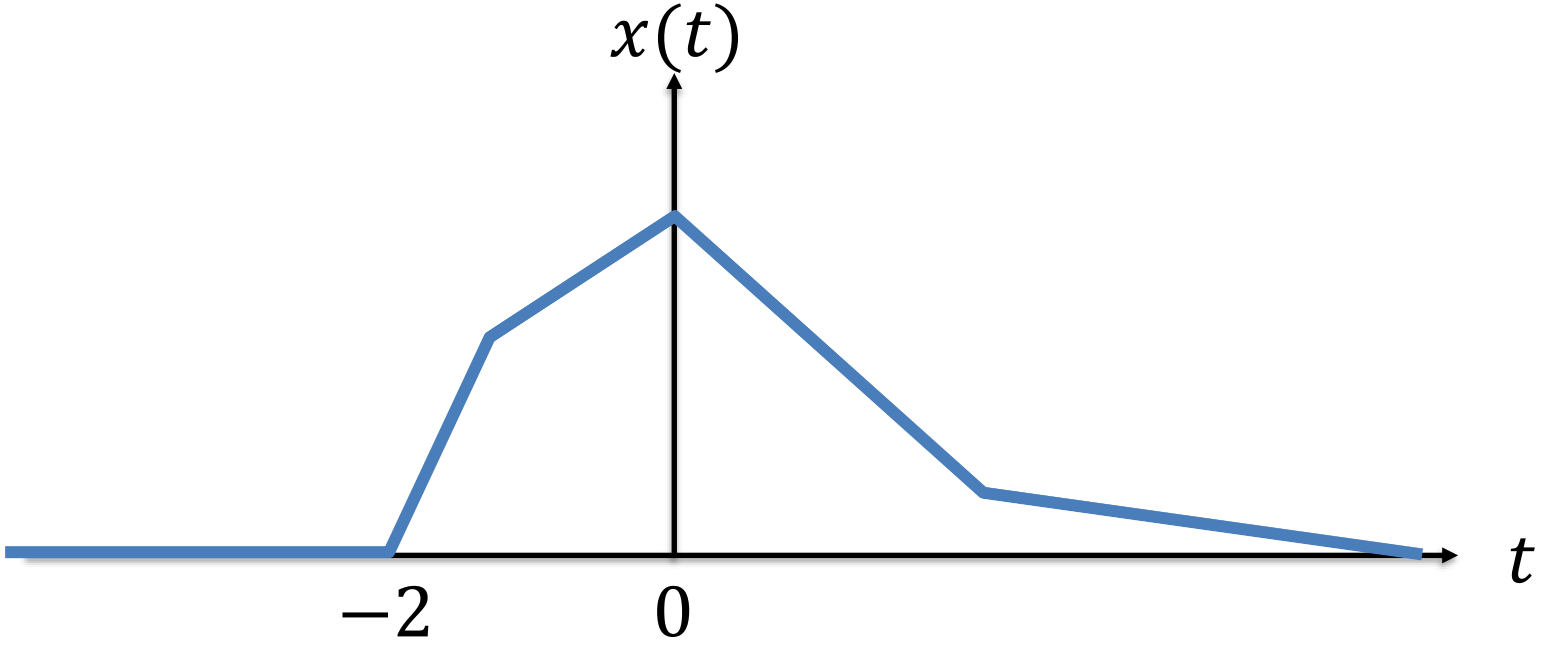
Property	$x(t)$		$\hat{\mathbf{X}}(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$
1. Multiplication by a constant	$K\, x(t)$	$\longleftrightarrow$	$K\, \hat{\mathbf{X}}(\omega)$
2. Linearity	$K_1\, x_1(t) + K_2\, x_2(t)$	$\longleftrightarrow$	$K_1\, \hat{\mathbf{X}}_1(\omega) + K_2\, \hat{\mathbf{X}}_2(\omega)$
3. Time scaling	$x(at)$	$\longleftrightarrow$	$\frac{1}{\lvert a \rvert}\, \hat{\mathbf{X}}\left(\frac{\omega}{a}\right)$
4. Time shift	$x(t - t_0)$	$\longleftrightarrow$	$e^{-j\omega t_0}\, \hat{\mathbf{X}}(\omega)$
5. Frequency shift	$e^{j\omega_0 t}\, x(t)$	$\longleftrightarrow$	$\hat{\mathbf{X}}(\omega - \omega_0)$
6. Time 1st derivative	$x' = \frac{dx}{dt}$	$\longleftrightarrow$	$j\omega\, \hat{\mathbf{X}}(\omega)$
7. Time nth derivative	$\frac{d^n x}{dt^n}$	$\longleftrightarrow$	$(j\omega)^n\, \hat{\mathbf{X}}(\omega)$
8. Time integral	$\int_{-\infty}^{t} x(\tau)\, d\tau$	$\longleftrightarrow$	$\frac{\hat{\mathbf{X}}(\omega)}{j\omega} + \pi\, \hat{\mathbf{X}}(0)\, \delta(\omega)$, where $\hat{\mathbf{X}}(0) = \int_{-\infty}^{\infty} x(t)\, dt$
9. Frequency derivative	$t^n\, x(t)$	$\longleftrightarrow$	$(j)^n \frac{d^n \hat{\mathbf{X}}(\omega)}{d\omega^n}$
10. Modulation	$x(t)\, \cos \omega_0 t$	$\longleftrightarrow$	$\frac{1}{2}[\hat{\mathbf{X}}(\omega - \omega_0) + \hat{\mathbf{X}}(\omega + \omega_0)]$
11. Convolution in t	$x_1(t) * x_2(t)$	$\longleftrightarrow$	$\hat{\mathbf{X}}_1(\omega)\, \hat{\mathbf{X}}_2(\omega)$
12. Convolution in ω	$x_1(t)\, x_2(t)$	$\longleftrightarrow$	$\frac{1}{2\pi}\, \hat{\mathbf{X}}_1(\omega) * \hat{\mathbf{X}}_2(\omega)$
13. Conjugate symmetry			$\hat{\mathbf{X}}(-\omega) = \hat{\mathbf{X}}^*(\omega)$

Today

- We will cover Sections 5-11 and 5-12 today
  - Circuit analysis with the Fourier Transform
  - The Fourier Transform method can be used if there are no initial conditions on capacitors and inductors
- The method is very similar to the Laplace transform technique and is useful for circuits excited by non-periodic waveforms
  - Today, our emphasis is on non-causal signals
- We are skipping Sections 5-9, 5-10

Thought question

What method would you use to analyze the following signal?



Can you use Laplace transforms directly?

Can you use a signal transformation to convert $x(t)$ into a causal signal $x_2(t)$?

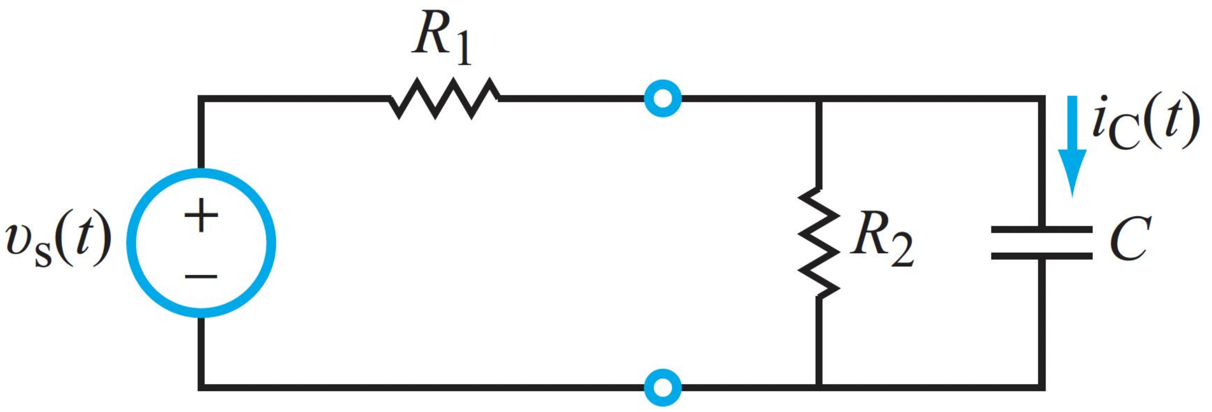
Can you use a Fourier transform?

Cheat sheet of Strategies

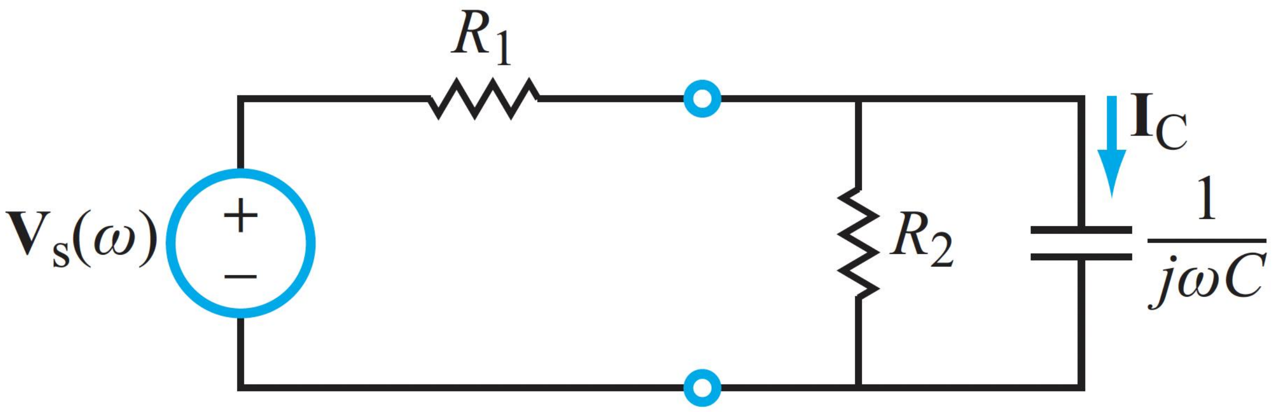
Input $x(t)$		Solution Method	Output $y(t)$
Duration	**Waveform**		
Everlasting	**Sinusoid**	Phasor Domain	Steady State Component (no transient exists)
Everlasting	**Periodic**	Phasor Domain and Fourier Series	Steady State Component (no transient exists)
Causal, $x(t) = 0$, for $t < 0$	**Any**	Laplace Transform (unilateral) (can accommodate non-zero initial conditions)	Complete Solution (transient + steady state)
Everlasting	**Any**	Bilateral Laplace Transform or Fourier Transform	Complete Solution (transient + steady state)

Example

Consider the following circuit



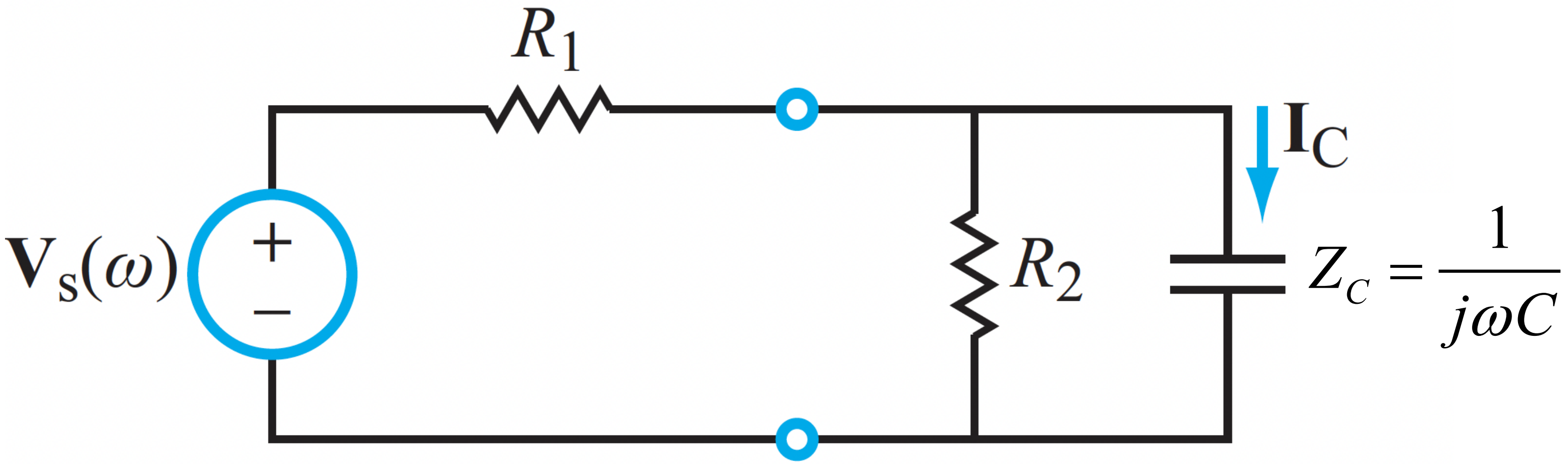
(a) Time domain



(b) ω-domain

SIMILAR TO LAPLACE TRANSFORM METHOD,
BUT NO INITIAL CONDITIONS

The Transfer Function



VOLTAGE DIVIDER

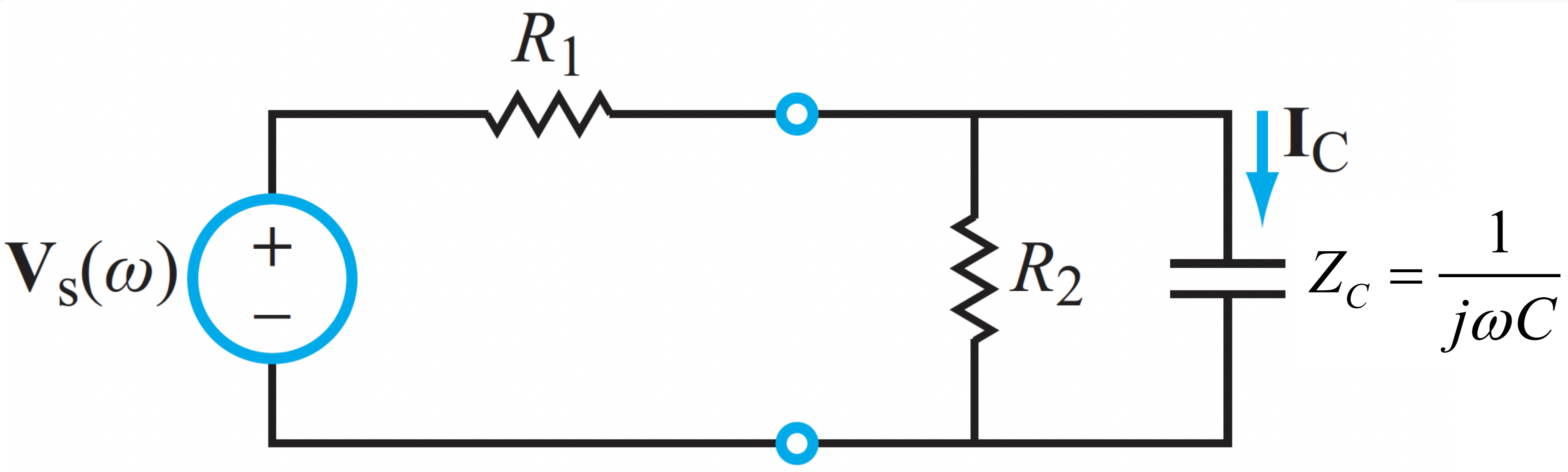
$$\hat{\mathbf{V}}_C(\omega) = \hat{\mathbf{V}}_S(\omega)\frac{R_2 \parallel Z_C}{R_1 + R_2 \parallel Z_C}$$

$$\hat{\mathbf{I}}_C(\omega) = \frac{\hat{\mathbf{V}}_C(\omega)}{Z_C} = \frac{1}{Z_C}\hat{\mathbf{V}}_S(\omega)\frac{R_2 \parallel Z_C}{R_1 + R_2 \parallel Z_C} = \frac{\hat{\mathbf{V}}_S(\omega)}{Z_C}\square\frac{R_2 Z_C / (R_2 + Z_C)}{R_1 + R_2 Z_C / (R_2 + Z_C)}$$

$$\hat{\mathbf{H}}(\omega) = \frac{\hat{\mathbf{I}}_C(\omega)}{\hat{\mathbf{V}}_S(\omega)} = \frac{j\omega / R_1}{\dfrac{R_1 + R_2}{R_1 R_2 C} + j\omega}$$

AFTER SIMPLIFYING

The Transfer Function



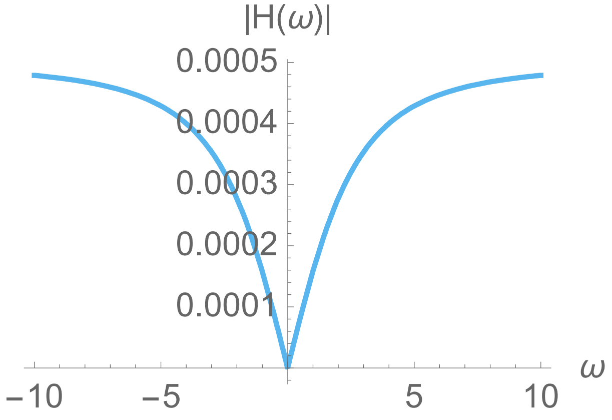
$$\hat{\mathbf{H}}(\omega) = \frac{\hat{\mathbf{I}}_C(\omega)}{\hat{\mathbf{V}}_S(\omega)} = \frac{j\omega / R_1}{\frac{R_1 + R_2}{R_1 R_2 C} + j\omega}$$

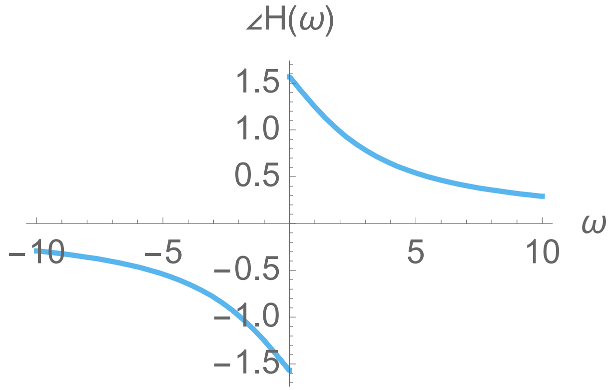
WE ARE NOW READY TO CALCULATE THE OUTPUT FOR A SPECIFIC INPUT

$$R_1 = 2\text{k}\Omega$$
$$R_2 = 4\text{k}\Omega$$
$$C = 0.25\text{mF}$$

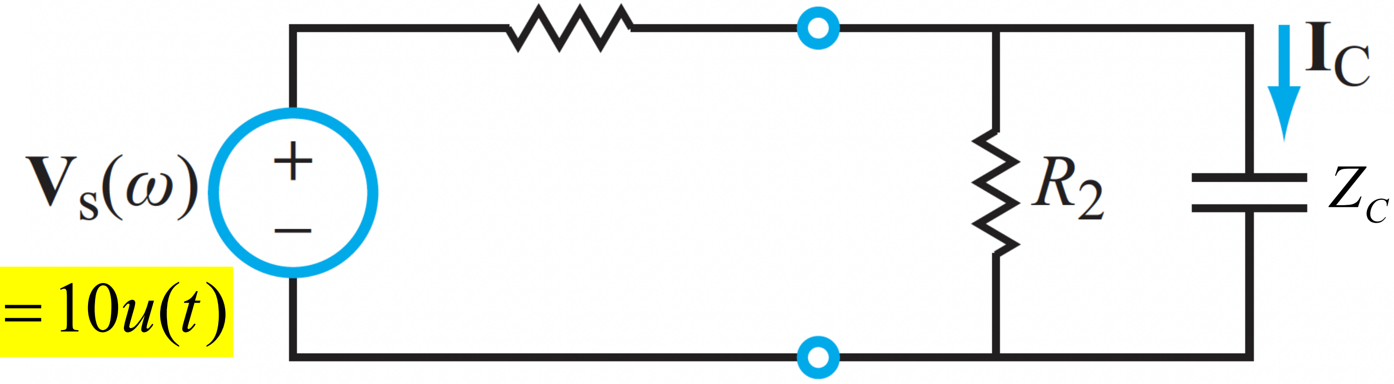
GIVEN VALUES

$$\Rightarrow \hat{\mathbf{H}}(\omega) = \frac{j\omega 0.5 \times 10^{-3}}{3 + j\omega}$$



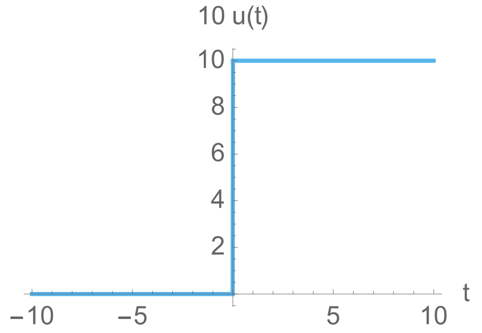


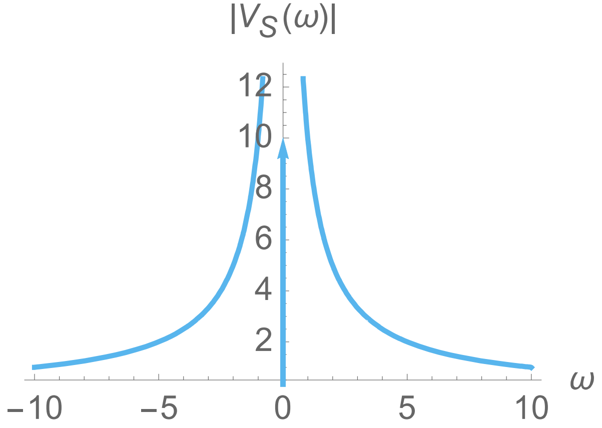
INPUT 1



$Z_C = \dfrac{1}{j\omega C}$

$\vartheta_S(t) = 10u(t)$





$$\hat{\mathbf{H}}(\omega) = \frac{j\omega 0.5\times 10^{-3}}{3+j\omega}$$

$$\vartheta_S(t) = 10u(t) \Rightarrow \hat{\mathbf{V}}_S(\omega) = 10\left[\pi\delta(\omega) + \frac{1}{j\omega}\right]$$

Entry #3 in Table 5-6

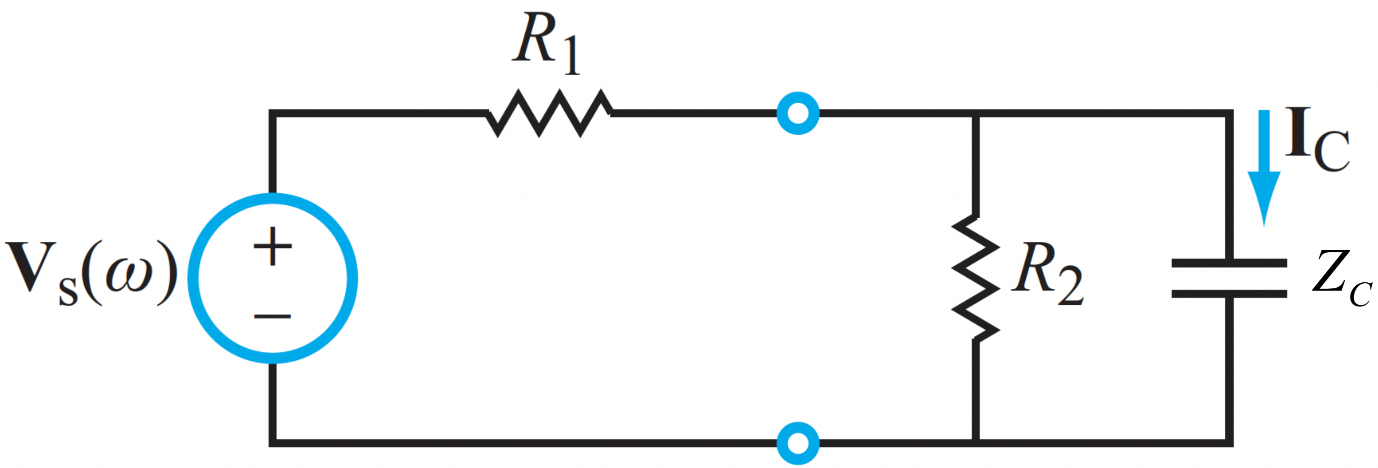
$$\Rightarrow \hat{\mathbf{I}}_C(\omega) = \hat{\mathbf{H}}(\omega)\hat{\mathbf{V}}_S(\omega)$$

$$= \frac{j\omega 0.5\times 10^{-3}}{3+j\omega}\times 10\left[\pi\delta(\omega)+\frac{1}{j\omega}\right] = 5\times 10^{-3}\times\left[\frac{j\omega\pi\delta(\omega)}{3+j\omega}+\frac{1}{3+j\omega}\right]$$

NEED TO INVERT THIS TO TIME DOMAIN

INPUT 1

$$\vartheta_S(t) = 10u(t)$$



$$Z_C = \frac{1}{j\omega C}$$

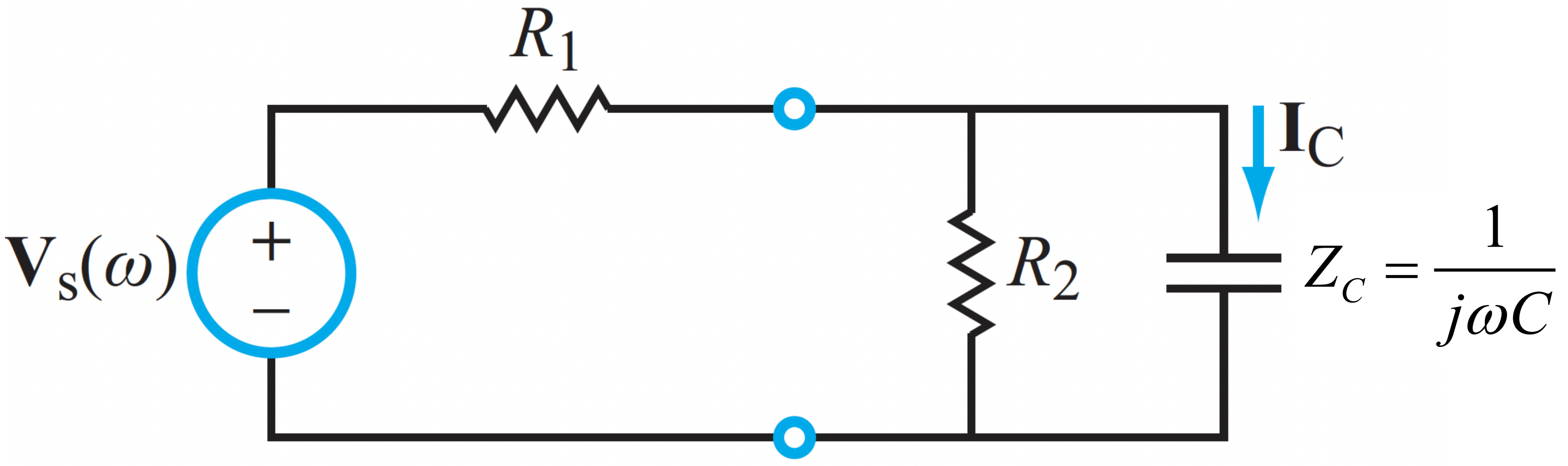
$$\hat{\mathbf{I}}_C(\omega) = 5\pi \square 10^{-3} \square \frac{j\omega\delta(\omega)}{3+j\omega} + 5 \square 10^{-3} \square \frac{1}{3+j\omega}$$

NO TABLE ENTRY
NEED TO WORK
FROM DEFINITION
(NEXT)

ENTRY #7 in
TABLE 5 - 6

$$\boxed{\mathcal{F}^{-1}\left[\frac{1}{3+j\omega}\right] = e^{-3t}u(t)}$$

INPUT 1



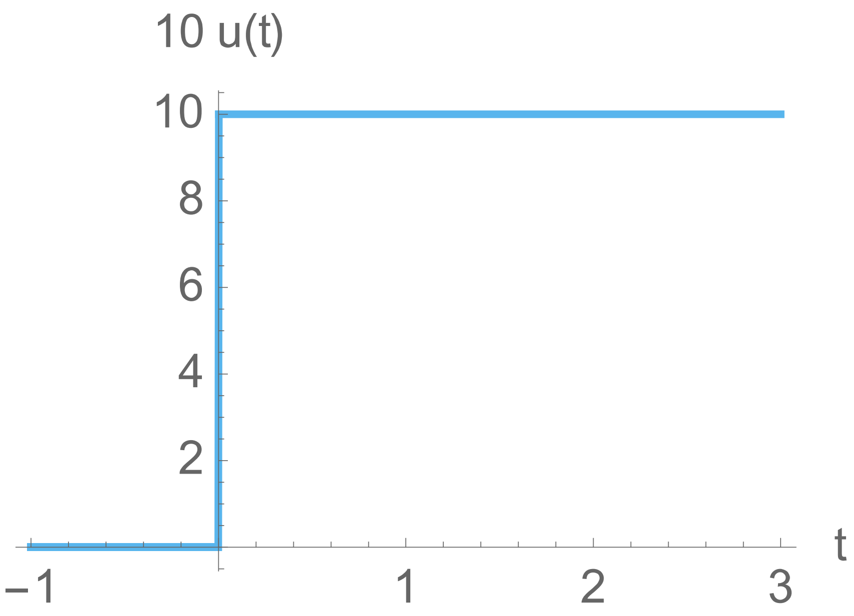
$$\hat{\mathbf{I}}_C(\omega) = 5\pi \square 10^{-3} \square \frac{j\omega\delta(\omega)}{3+j\omega} + 5 \square 10^{-3} \square \frac{1}{3+j\omega}$$

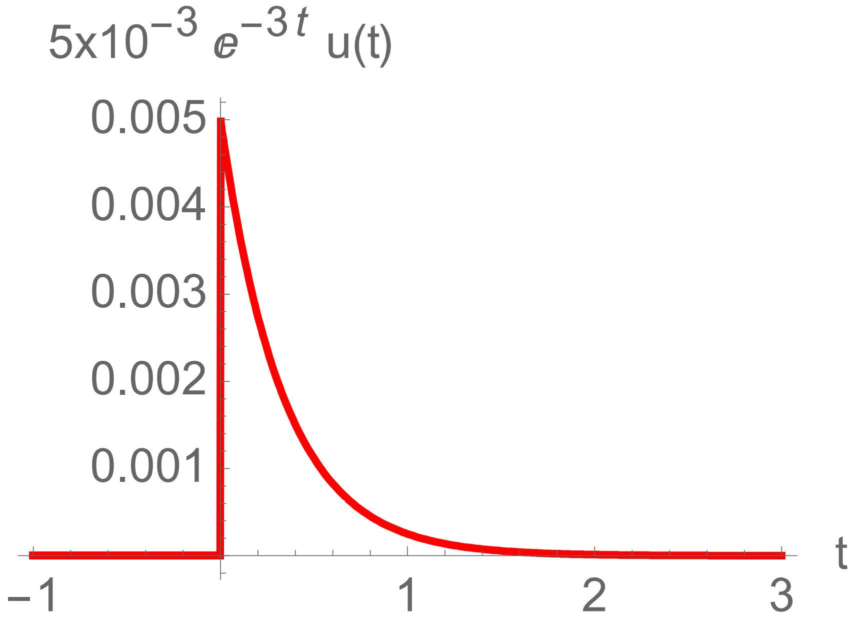
$$\boxed{\mathcal{F}^{-1}\left[\frac{j\omega\delta(\omega)}{3+j\omega}\right] = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\frac{j\omega\delta(\omega)}{3+j\omega}\right]e^{j\omega t}\,d\omega = 0}$$

SAMPLING PROPERTY OF THE IMPULSE

$$\square \quad i_C(t) = 5 \square 10^{-3} e^{-3t} u(t)$$

PLOT OF INPUT 1

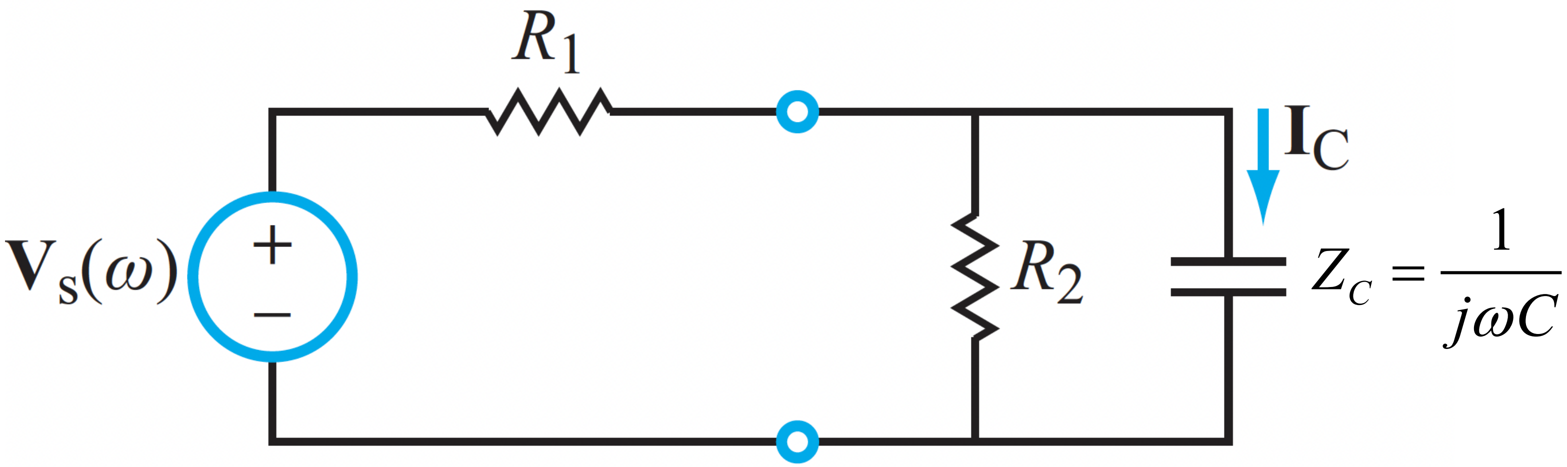


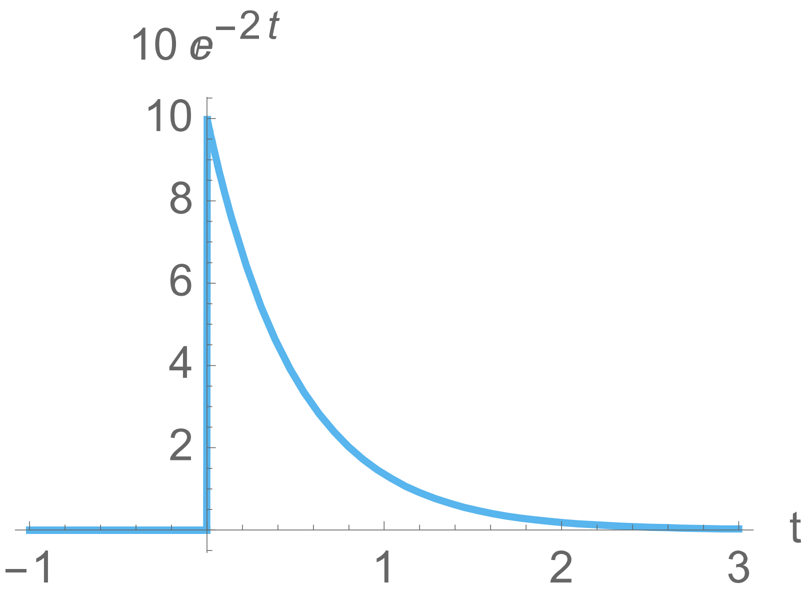


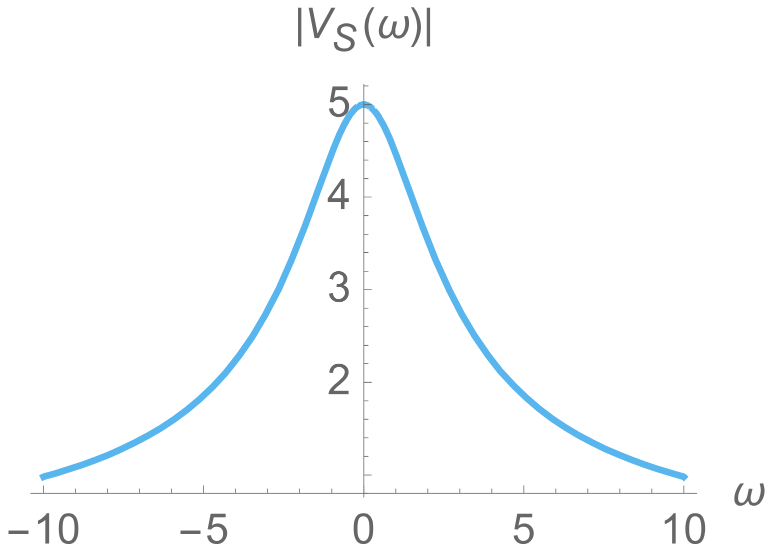
□ $i_C(t) = 5 \square 10^{-3} e^{-3t} u(t)$

INPUT 2

$$\vartheta_S(t) = 10e^{-2t}u(t)$$

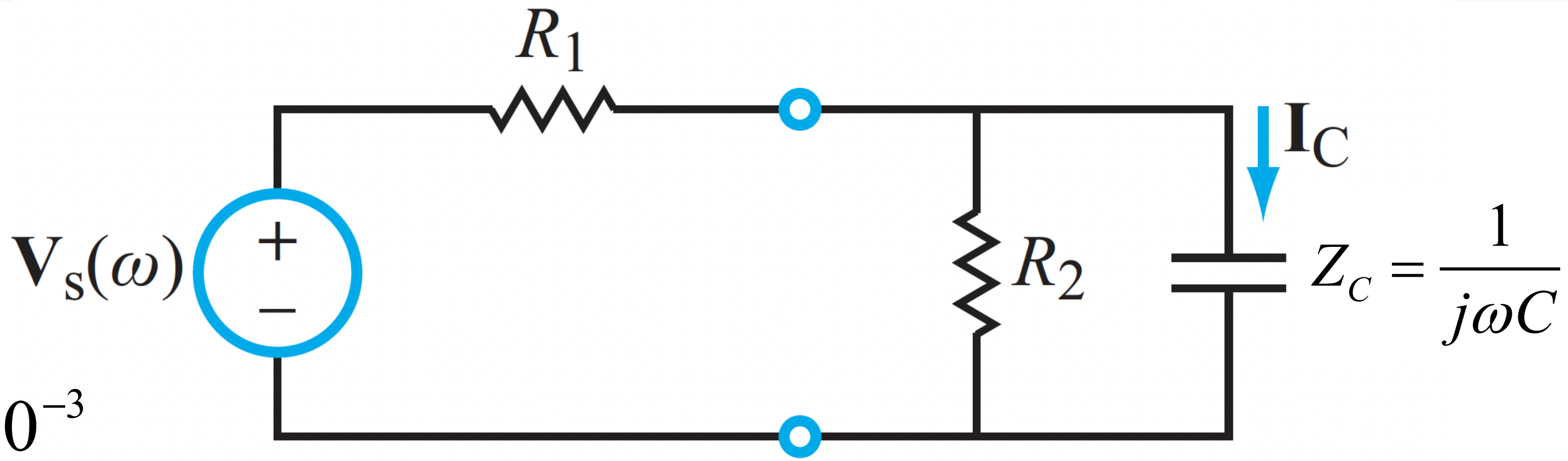






INPUT 2

$$\vartheta_S(t) = 10e^{-2t}u(t)$$



$$\hat{\mathbf{H}}(\omega) = \frac{j\omega 0.5\times 10^{-3}}{(3+j\omega)}$$

SAME AS BEFORE

$$\vartheta_S(t) = 10e^{-2t}u(t) \Rightarrow \hat{\mathbf{V}}_S(\omega) = \frac{10}{(2+j\omega)}$$

Entry #7 in Table 5-6

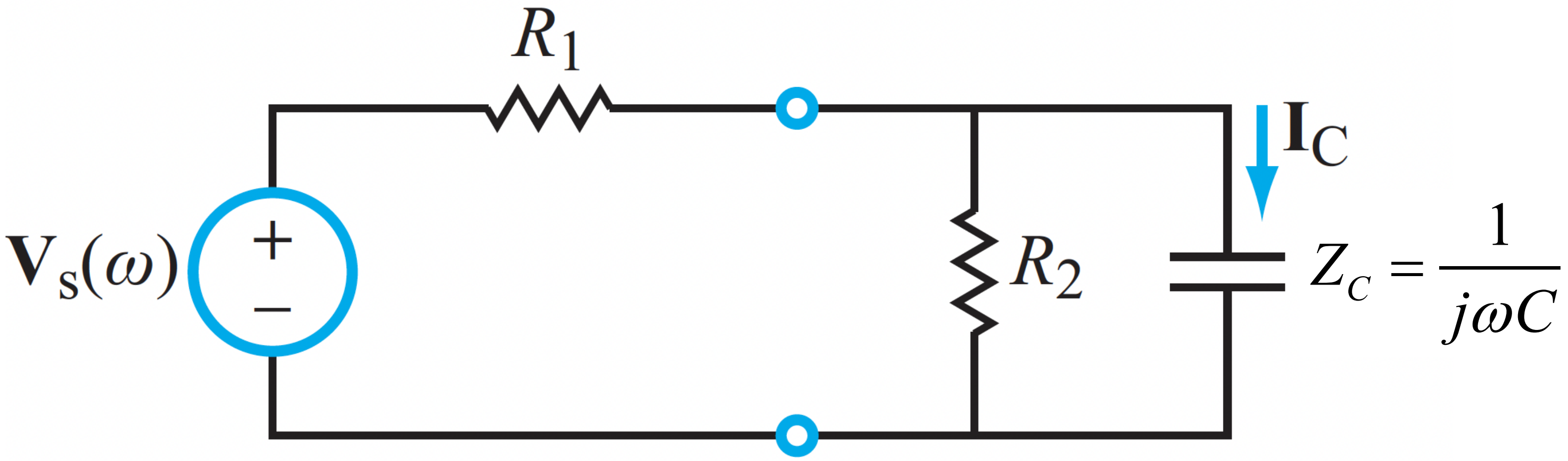
$$\Rightarrow \hat{\mathbf{I}}_C(\omega) = \hat{\mathbf{H}}(\omega)\hat{\mathbf{V}}_S(\omega)$$

$$= \frac{j\omega 0.5\times 10^{-3}}{(3+j\omega)} \times \frac{10}{(2+j\omega)} = 5\times 10^{-3}\left[\frac{3}{(3+j\omega)} - \frac{2}{(2+j\omega)}\right]$$

PARTIAL FRACTION EXPANSION

INPUT 2

$$\vartheta_s(t) = 10e^{-2t}u(t)$$



$$\hat{\mathbf{I}}_C(\omega) = 5 \times 10^{-3}\left[\frac{3}{(3 + j\omega)} - \frac{2}{2 + j\omega}\right]$$

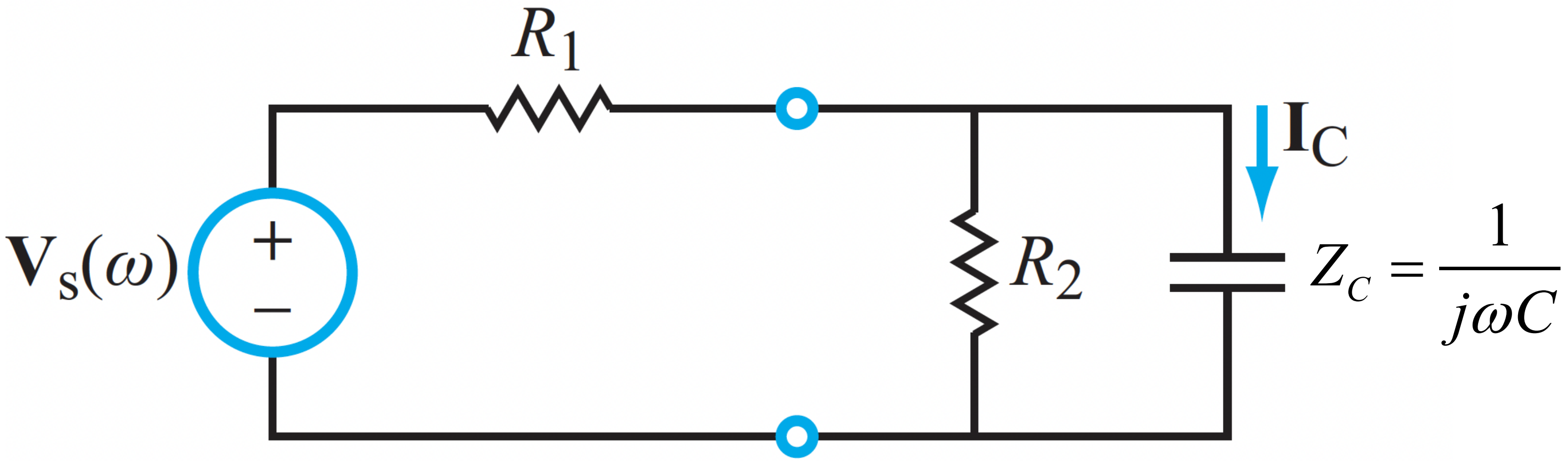
$$\Rightarrow i_C(t) = 5 \times 10^{-3}\left[3e^{-3t} - 2e^{-2t}\right]u(t)$$

$$= (15e^{-3t} - 10e^{-2t})u(t) \ \ \text{mA}$$

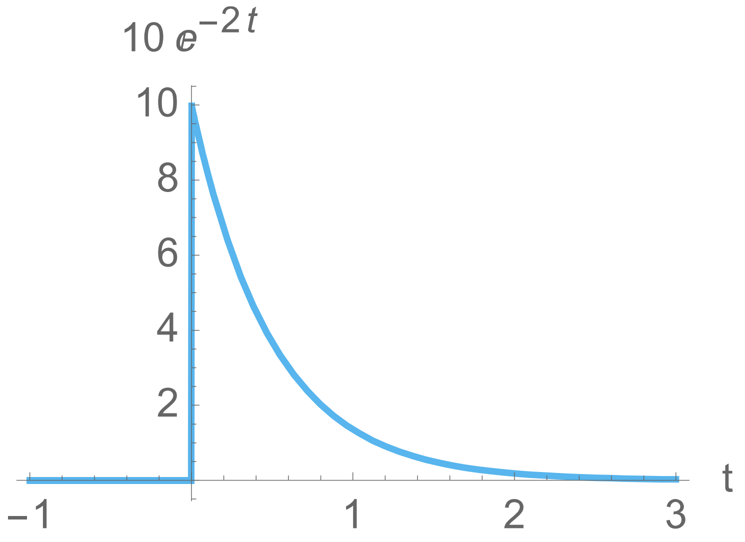
Entry #7
in Table 5-6

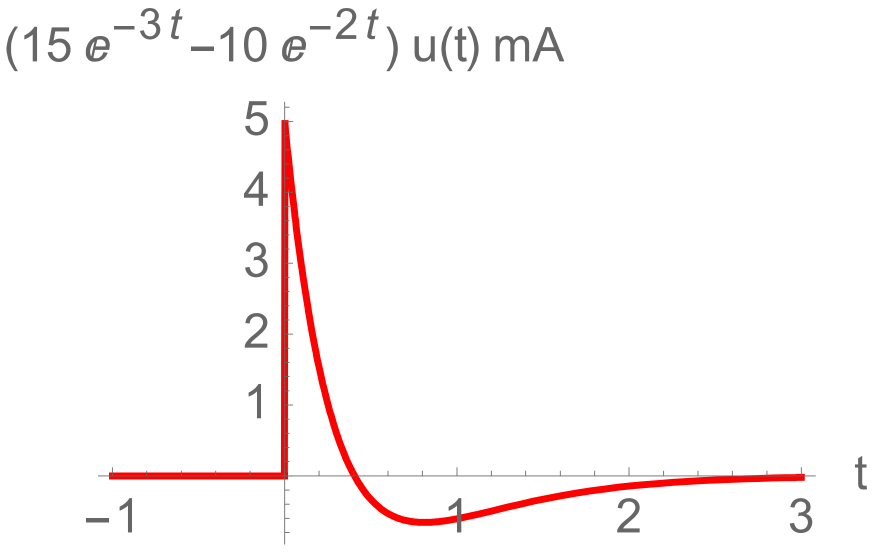
INPUT 2

$$\vartheta_S(t) = 10e^{-2t}u(t)$$



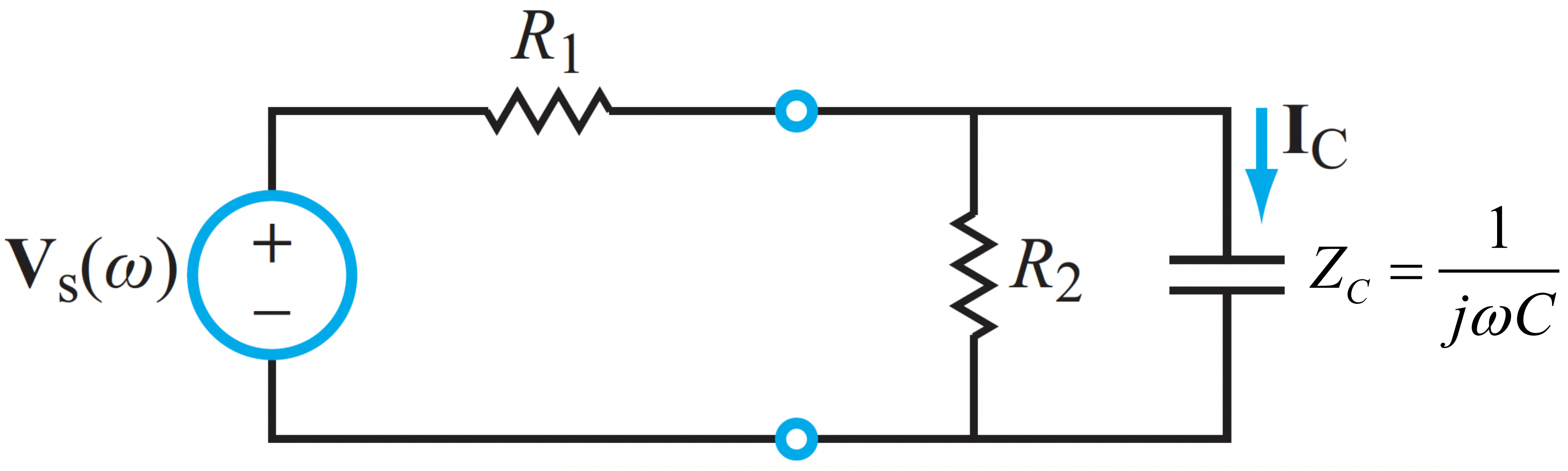
$$i_C(t) = (15e^{-3t} - 10e^{-2t})u(t) \text{ mA}$$



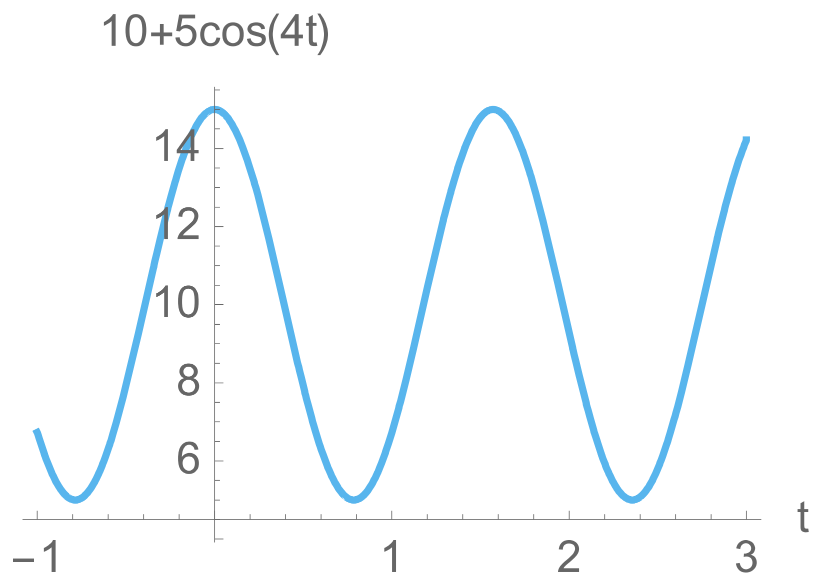


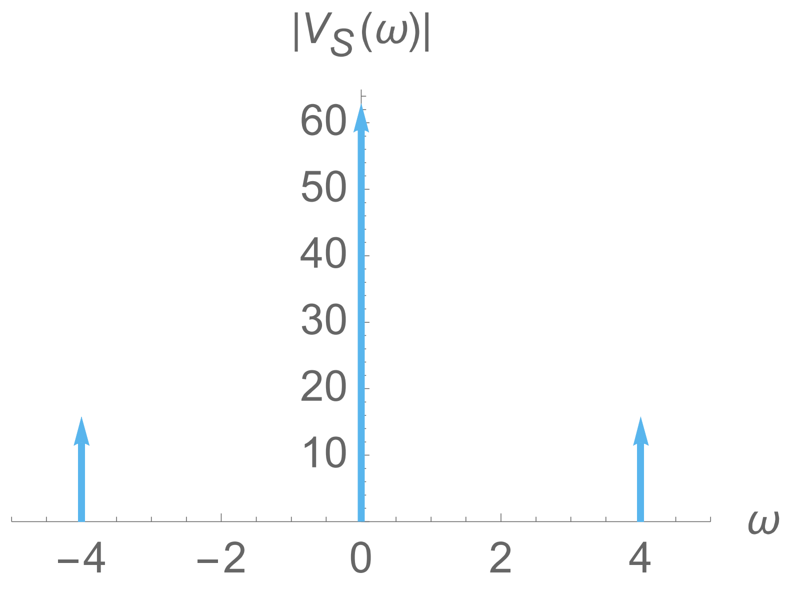
INPUT 3

$$\vartheta_s(t) = 10 + 5\cos 4t$$



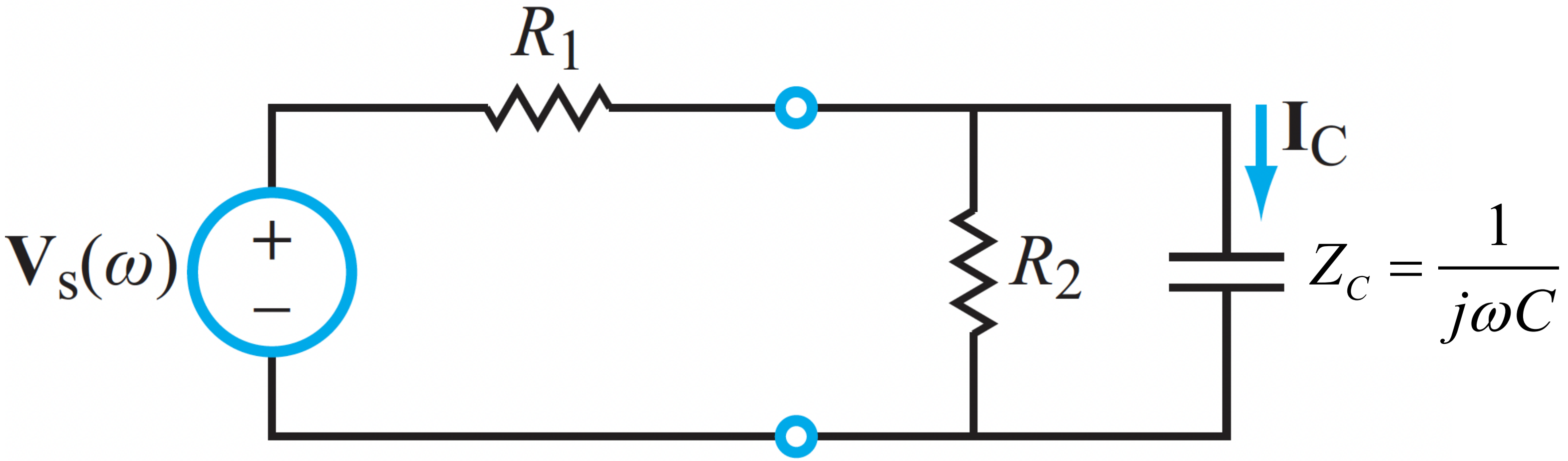
$$\hat{\mathbf{V}}_s(\omega) = 20\pi\delta(\omega) + 5\pi\left[\delta(\omega-4) + \delta(\omega+4)\right]$$





INPUT 3

$$\vartheta_S(t) = 10 + 5\cos 4t$$



$$\hat{\mathbf{V}}_S(\omega) = 20\pi\delta(\omega) + 5\pi\left[\delta(\omega-4)+\delta(\omega+4)\right]$$

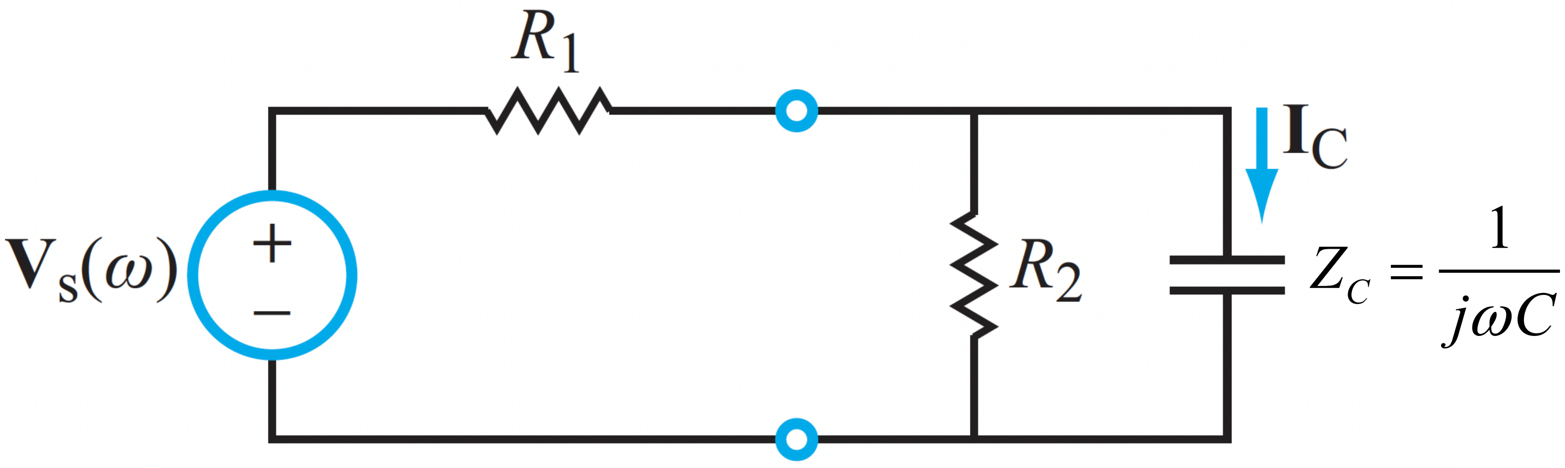
$$\Rightarrow \hat{\mathbf{I}}_C(\omega) = \hat{\mathbf{H}}(\omega)\hat{\mathbf{V}}_S(\omega) = \frac{j\omega 0.5\times 10^{-3}}{(3+j\omega)} \times \hat{\mathbf{V}}_S(\omega)$$

$$= \frac{j\pi\omega\delta(\omega)10\times 10^{-3}}{(3+j\omega)} + j\pi 2.5\times 10^{-3}\left[\frac{\omega\delta(\omega-4)}{(3+j\omega)} + \frac{\omega\delta(\omega+4)}{(3+j\omega)}\right]$$

NO TABLE ENTRY. NEED TO WORK FROM BASIC DEFINITION

INPUT 3

$$\vartheta_S(t) = 10 + 5\cos 4t$$



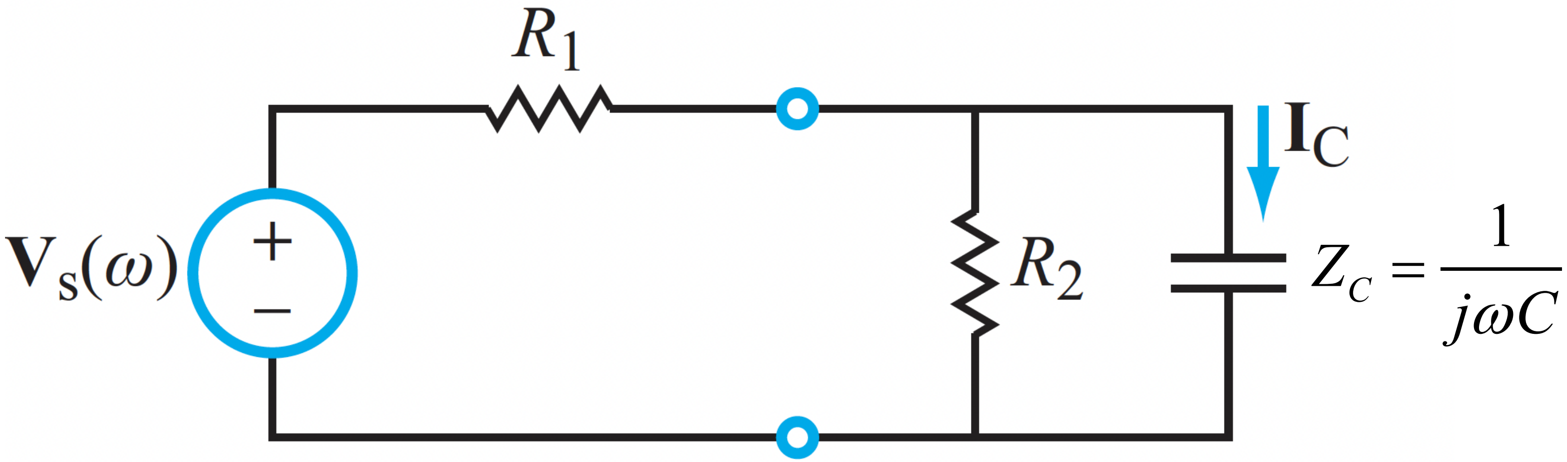
$$\hat{\mathbf{I}}_C(\omega) = \frac{j\pi\omega 10\times 10^{-3}}{(3+j\omega)}\delta(\omega) + j\pi 2.5\times 10^{-3}\left[\frac{\omega\delta(\omega-4)}{(3+j\omega)} + \frac{\omega\delta(\omega+4)}{(3+j\omega)}\right]$$

WE CAN AGAIN TAKE ADVANTAGE OF THE SAMPLING PROPERTY

$$\boxed{i_C(t) = \mathcal{F}^{-1}\left[\hat{\mathbf{I}}_C(\omega)\right] = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\hat{\mathbf{I}}_C(\omega)\right]e^{j\omega t}\,d\omega}$$

INPUT 3

$$\vartheta_S(t) = 10 + 5\cos 4t$$



$$\hat{\mathbf{I}}_C(\omega) = \frac{j\pi\omega 10\times10^{-3}}{(3+j\omega)}\delta(\omega) + j\pi 2.5\times10^{-3}\left[\frac{\omega\delta(\omega-4)}{(3+j\omega)} + \frac{\omega\delta(\omega+4)}{(3+j\omega)}\right]$$

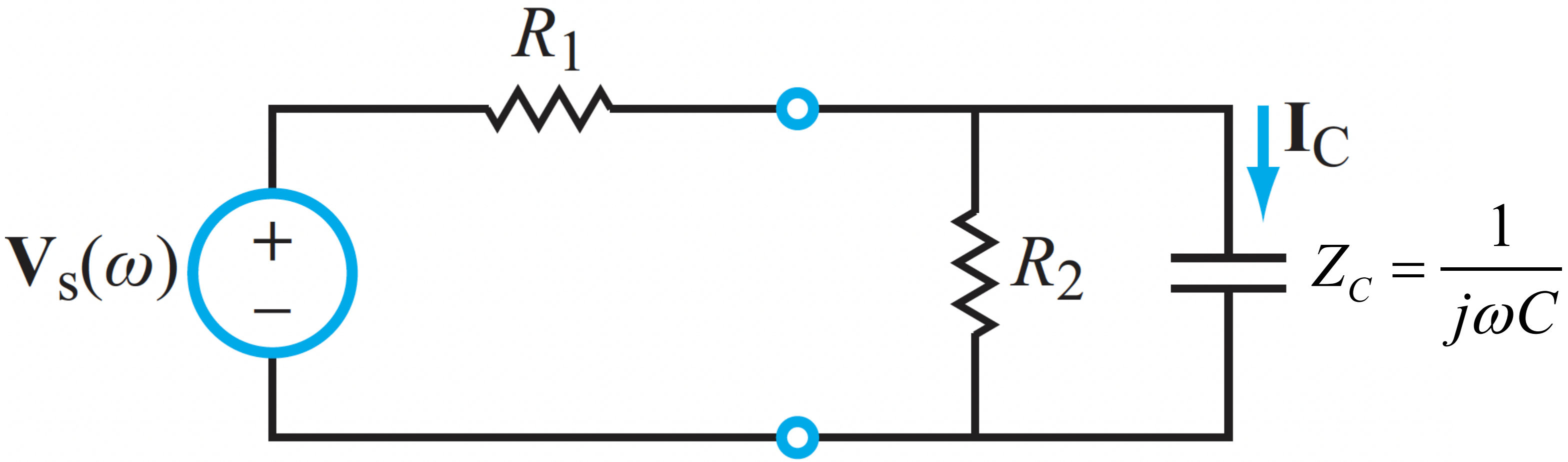
$$i_C(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\frac{j\pi\omega 10\times10^{-3}}{(3+j\omega)}\delta(\omega)\right]e^{j\omega t}d\omega + \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[j\pi 2.5\times10^{-3}\frac{\omega\delta(\omega-4)}{(3+j\omega)}\right]e^{j\omega t}d\omega$$

$$+\frac{1}{2\pi}\int_{-\infty}^{\infty}\left[j\pi 2.5\times10^{-3}\frac{\omega\delta(\omega+4)}{(3+j\omega)}\right]e^{j\omega t}d\omega$$

$$= 0 + \frac{j5\times10^{-3}e^{j4t}}{(3+j4)} - \frac{j5\times10^{-3}e^{-j4t}}{(3-j4)}$$

INPUT 3

$$\vartheta_S(t) = 10 + 5\cos 4t$$

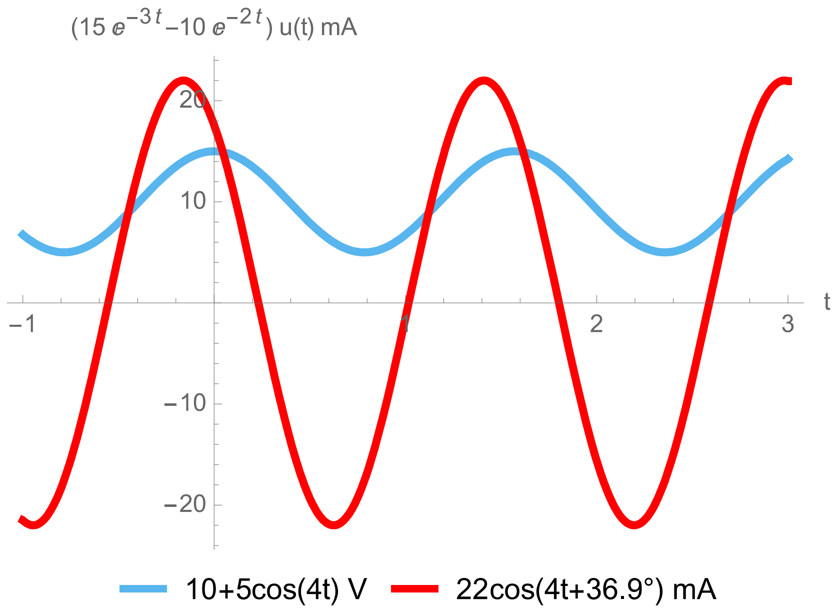


$$i_C(t) = 0 + \frac{j5\times10^{-3}e^{j4t}}{(3+j4)} + \frac{(-j)5\times10^{-3}e^{-j4t}}{(3-j4)}$$

$$= 5\times10^{-3}\left(\frac{e^{j90^o}e^{j4t}}{5e^{j53^o}} + \frac{e^{-j90^o}e^{-j4t}}{5e^{-j53^o}}\right)$$

$$= 5\times10^{-3}\left(\frac{e^{j4t}e^{j37^o}}{5} + \frac{e^{-j4t}e^{-j37^o}}{5}\right)$$

$$= 2\times10^{-3}\cos(4t+37^o)$$



WATCH: DIFFERENT SCALES

Now the Fun Stuff: Signal Filters

$$\mathbf{X}(\omega) \rightarrow \boxed{\begin{array}{c}\text{LTI System}\\ \mathbf{H}(\omega)\end{array}} \rightarrow \mathbf{Y}(\omega)$$

Filter

LTI systems can be renamed as signal "filters"

- Filters modify the amplitude and phase of signals in a frequency-dependent manner

Signal Filters are used widely!

- Radio, television, telephones, cell phones, satellites, automotive components, industrial systems, music players, computers, multi-media systems, concert sound and lighting, movie making, home appliances, …

Gain and Phase Angle of a Filter

$$\mathbf{X}(\omega) \rightarrow \boxed{\begin{array}{c}\text{LTI System}\\ \mathbf{H}(\omega)\end{array}} \rightarrow \mathbf{Y}(\omega)$$

Filter

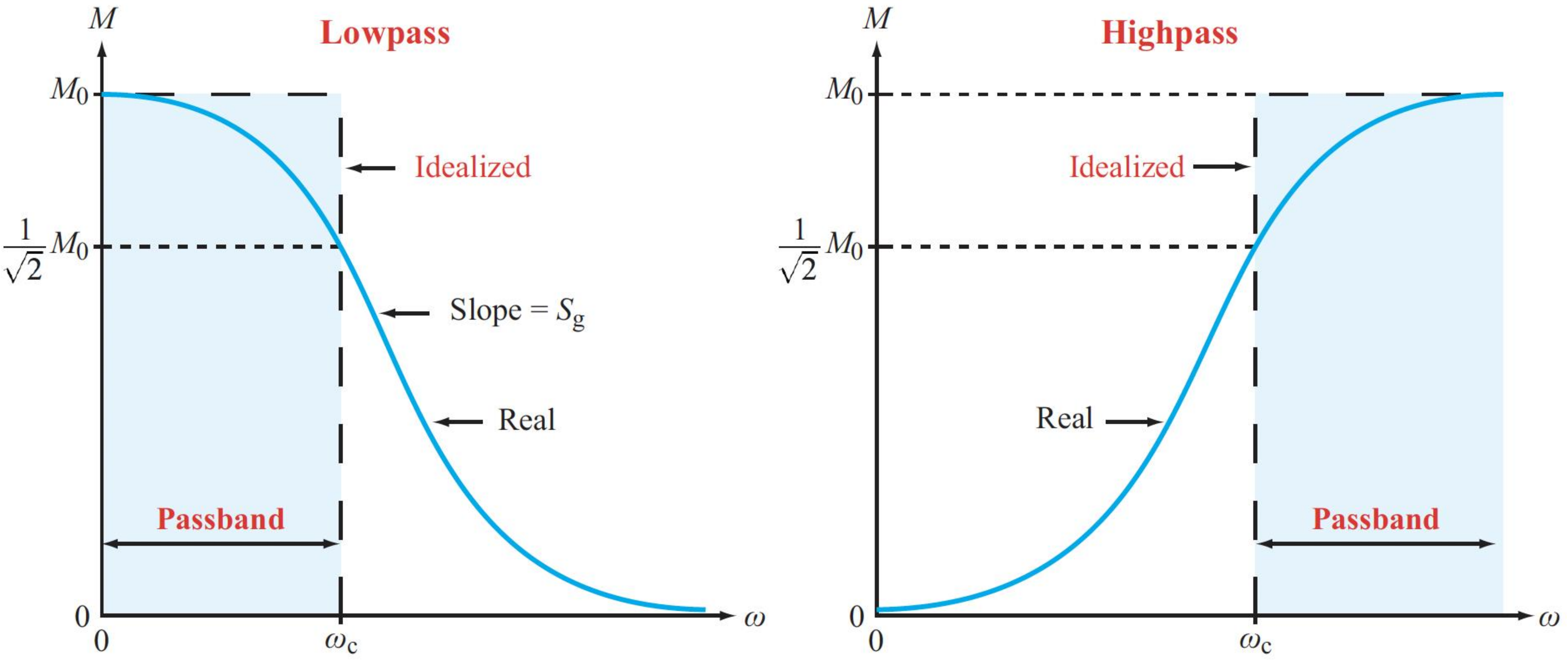
$$\hat{\mathbf{H}}(\omega) = \frac{\hat{\mathbf{X}}_{out}(\omega)}{\hat{\mathbf{X}}_{in}(\omega)} = \left|\hat{\mathbf{H}}(\omega)\right| e^{j\phi(\omega)} = M(\omega)e^{j\phi(\omega)}$$

GAIN

PHASE ANGLE

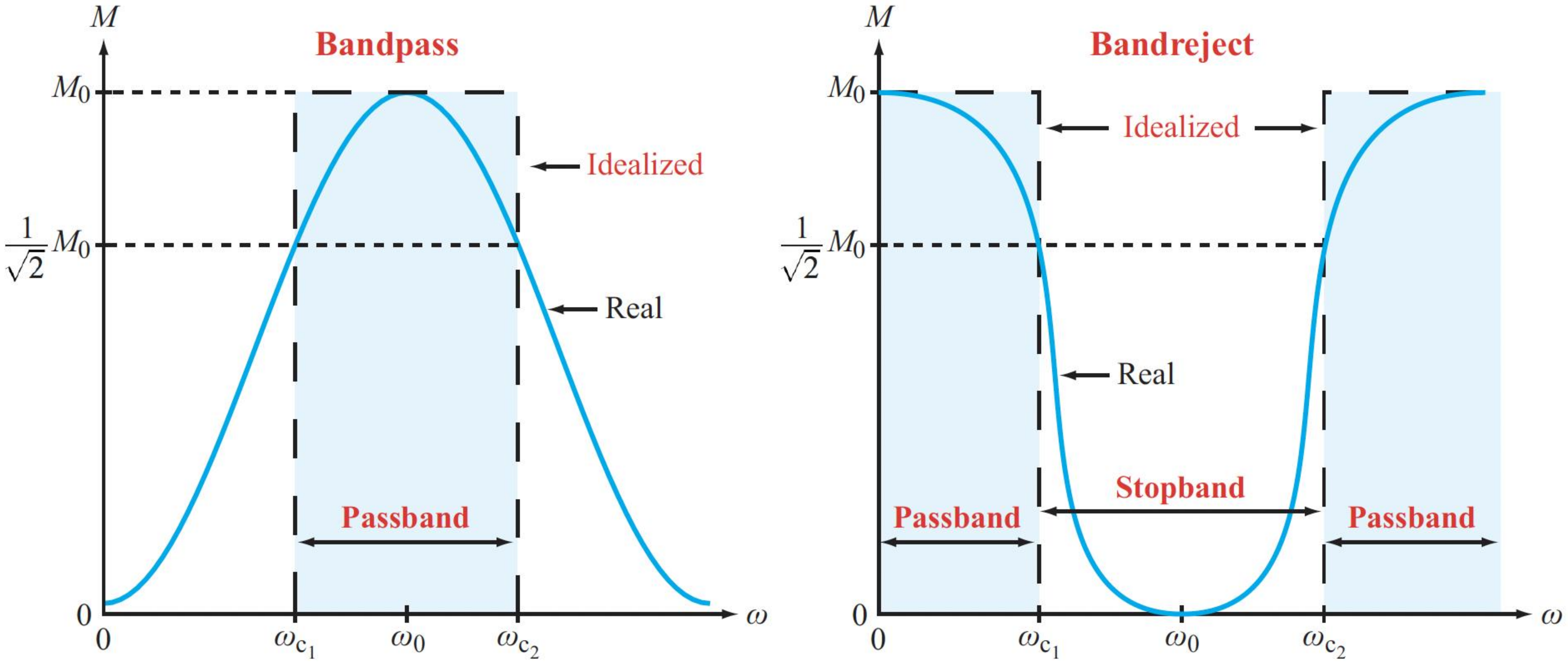
SAME QUANTITIES, NEW NAMES

Common Basic Filters



Can you think of good use cases?

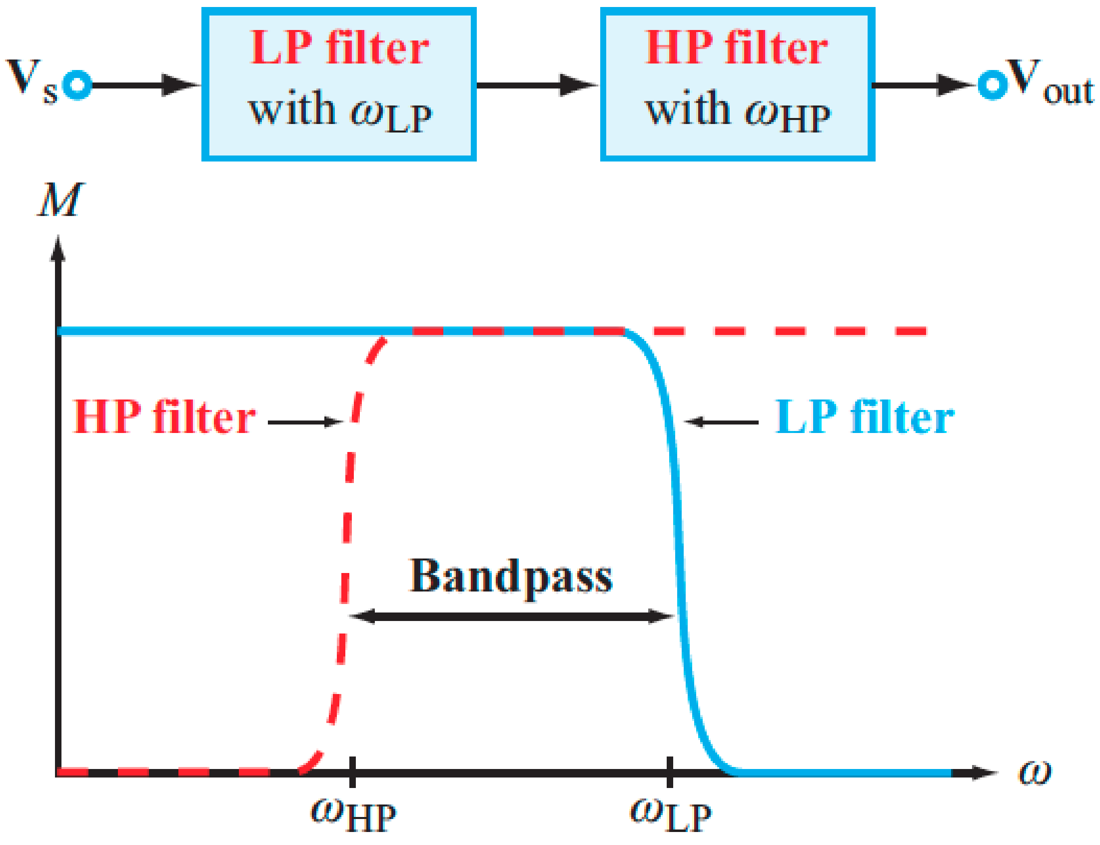
Common Basic Filters



Can you think of good use cases?

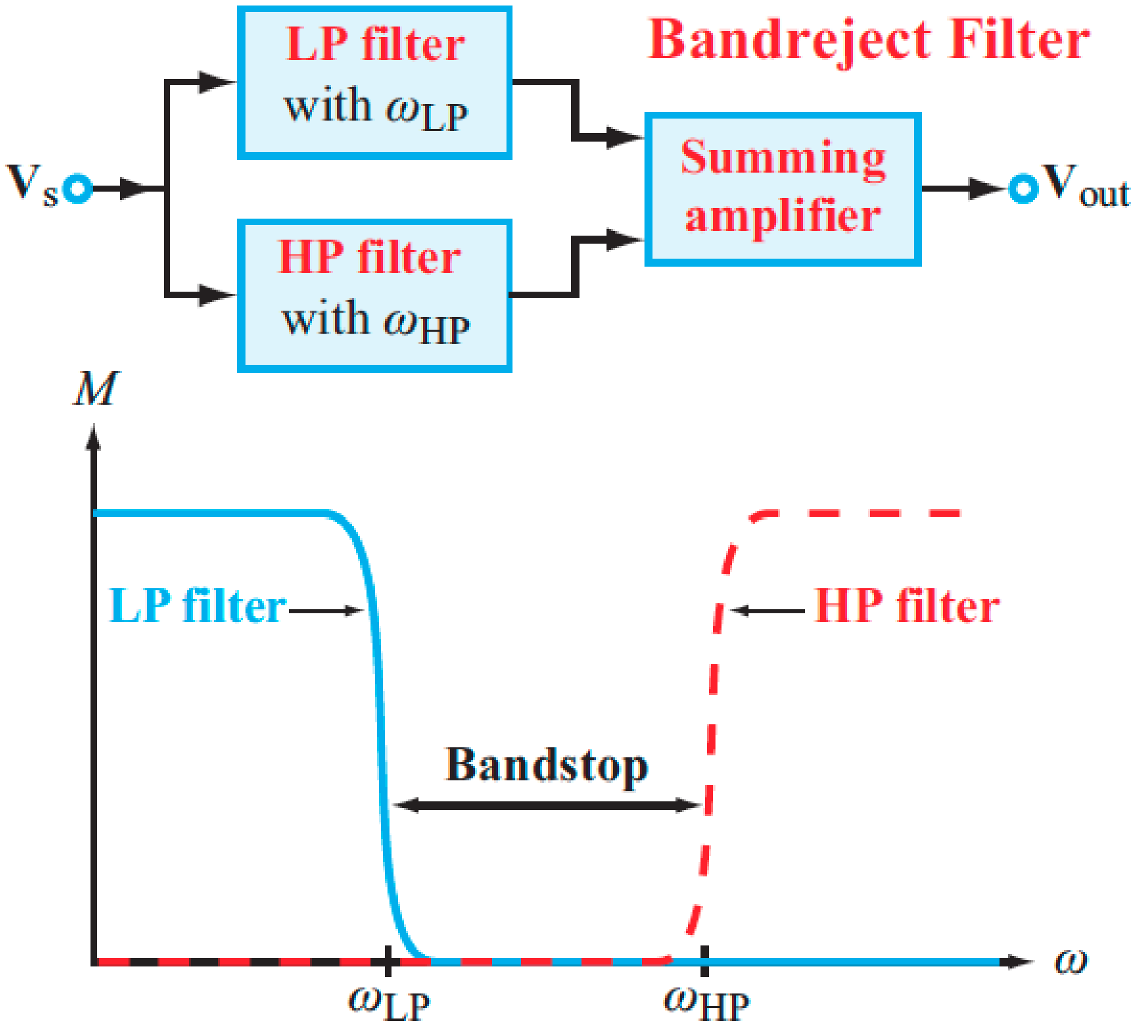
Combining Filters

Bandpass Filter



(a) Bandpass filter

Bandreject Filter



(b) Bandreject filter

Example 1

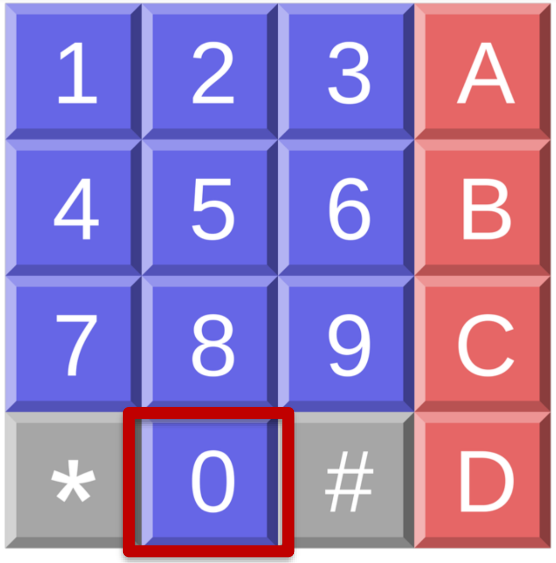
- A car rolls by, with windows rolled up, playing music with a heavy beat
  - From the outside, you mostly hear the thump of the beat
  - What kind of filtering are the car’s windows performing?

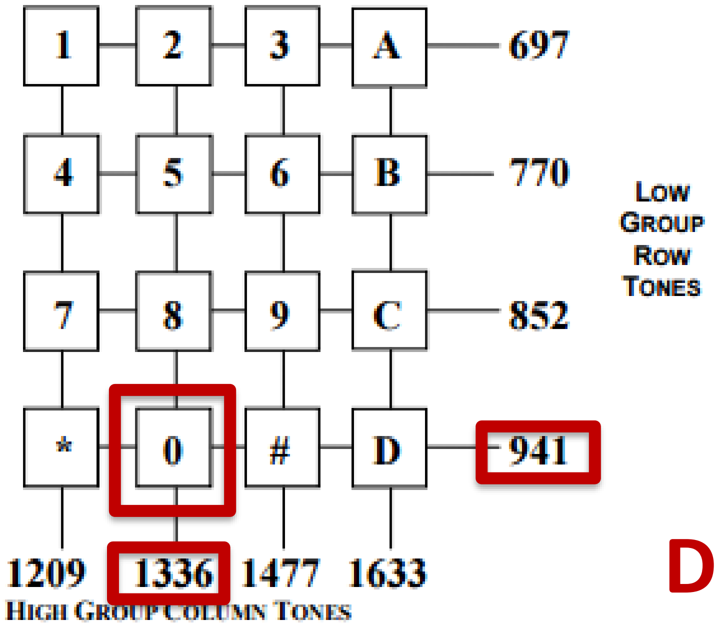
- You were trying to record a voice message when the above-mentioned car rolled by
  - What kind of filter do you need to apply to your recording to suppress the unwanted thumping sounds?

Grandma's Phone

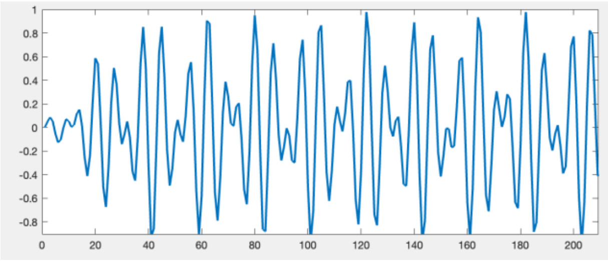
Mom’s Phone

How it Works

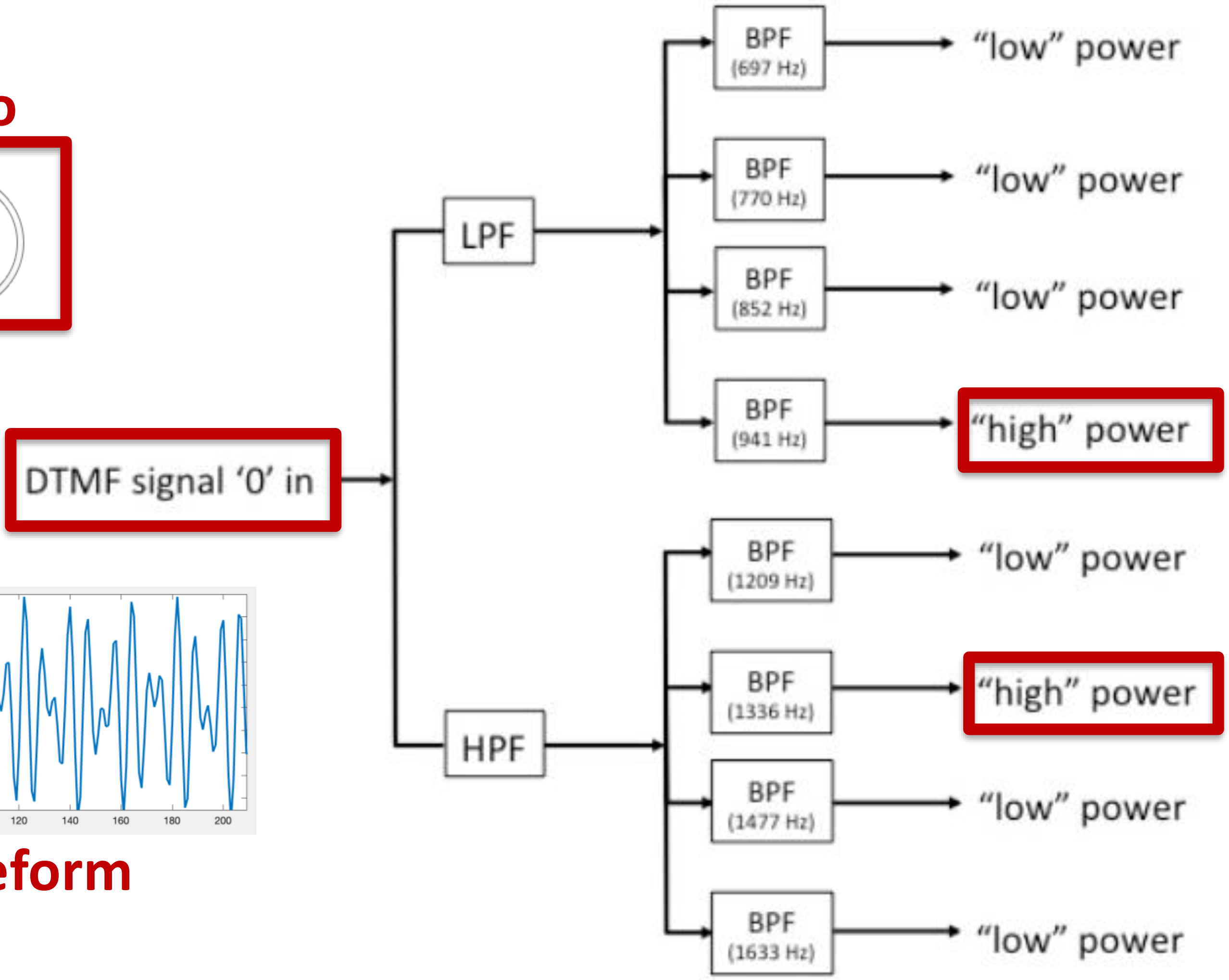




Audio



Waveform



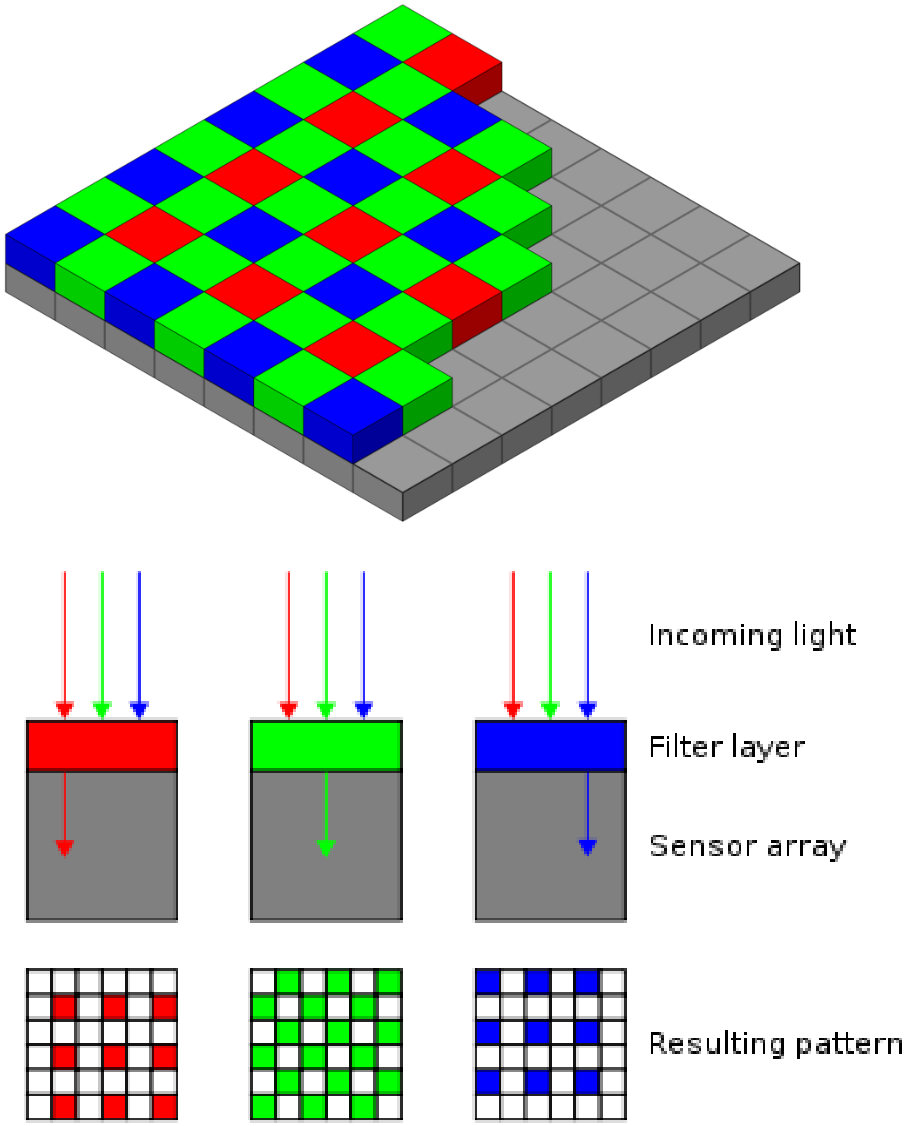
Dual-tone multi-frequency signaling (DTMF)

Graphic Audio Equalizer

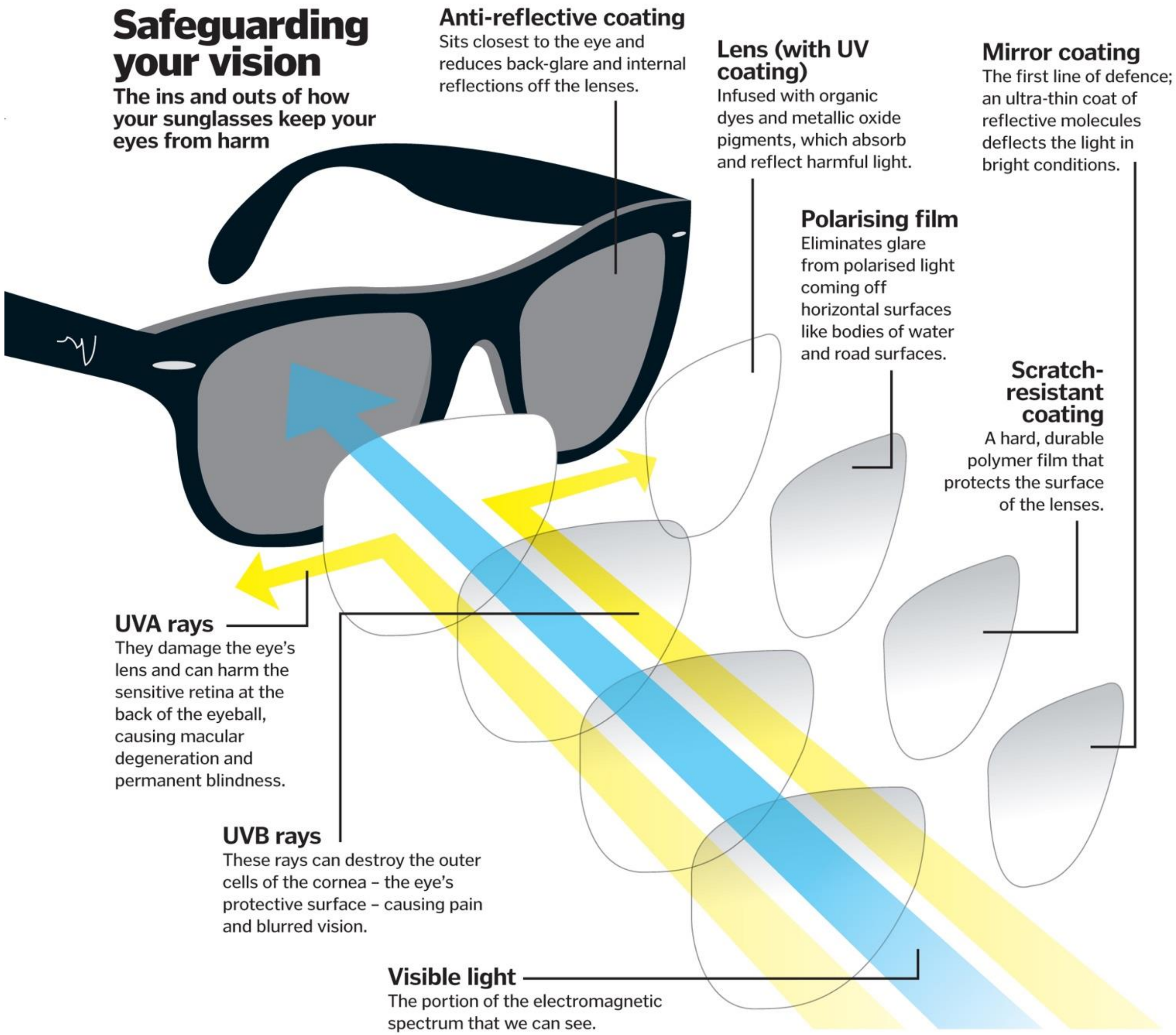
Allows you to shape the spectrum of an audio signal

- Each slider represents a band-pass filter. It can be used to either boost or suppress chosen frequencies
- The range of human hearing is 20Hz – 20kHz
- Sounds below 20Hz are called infrasound (elephants, earthquakes)
- Rhythm frequencies are in the 32 – 512Hz range
- The human voice is in the 300 – 3400Hz range
- Sounds above 20kHz are called ultrasound (bats and dogs)

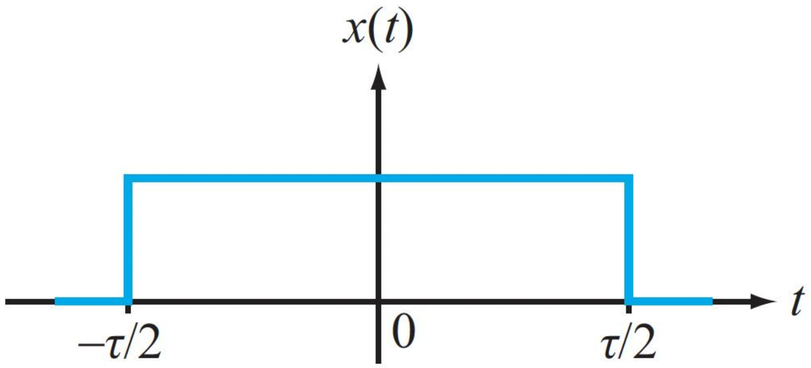
Optical Filters

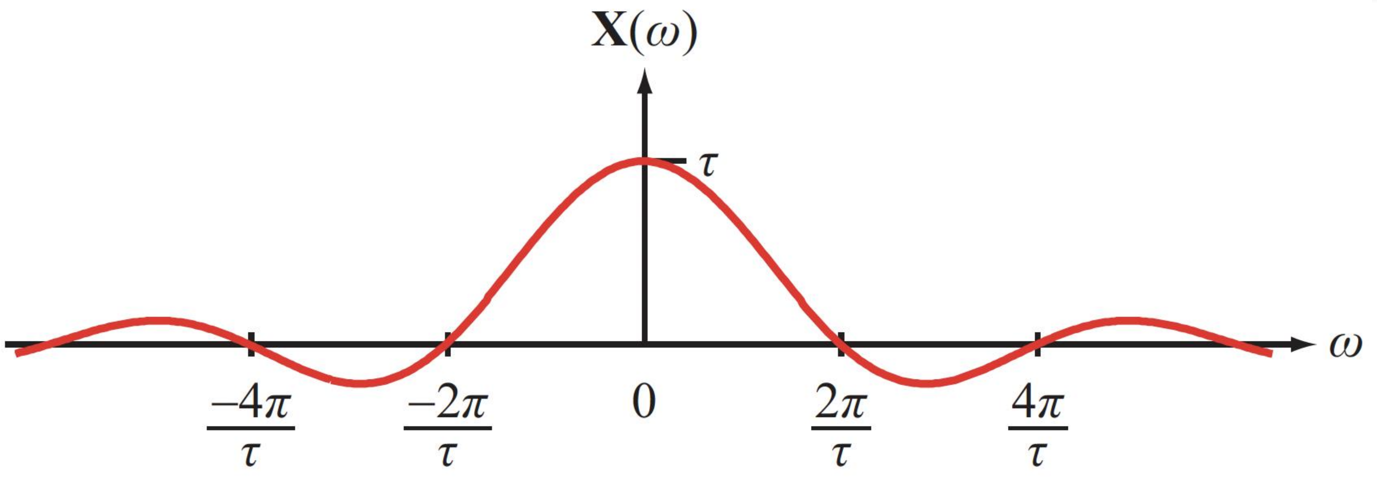


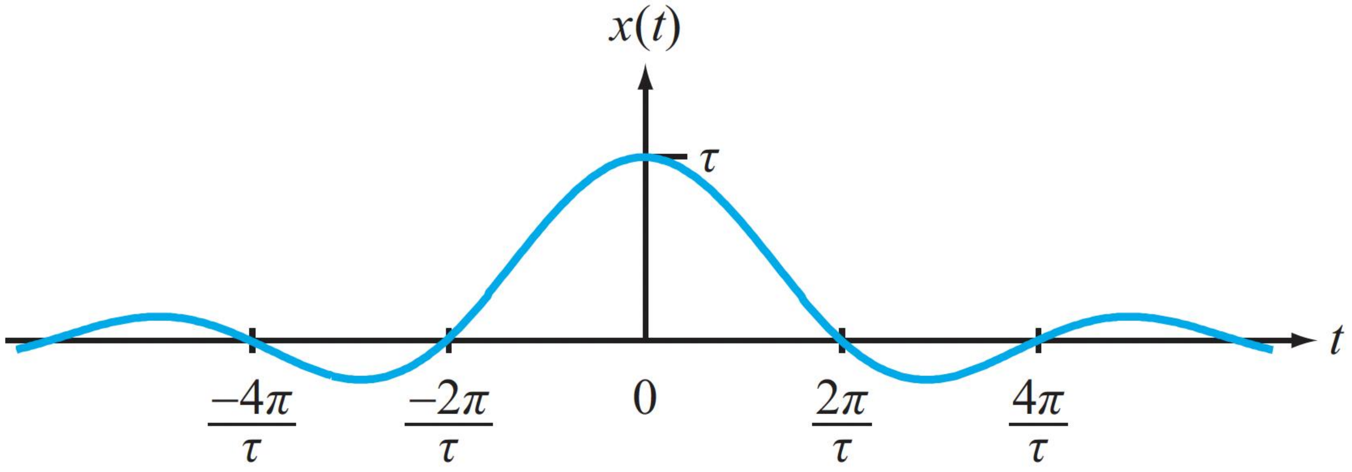
HOW COLOR VIDEO CAMERAS WORK

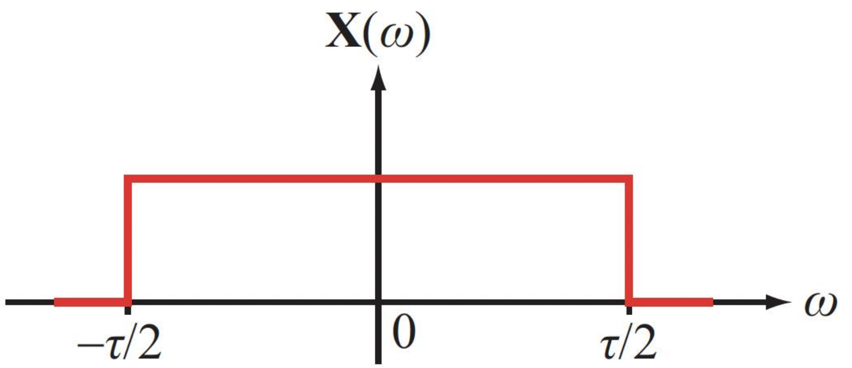


The Pulse $\leftrightarrow$ Fourier Transform



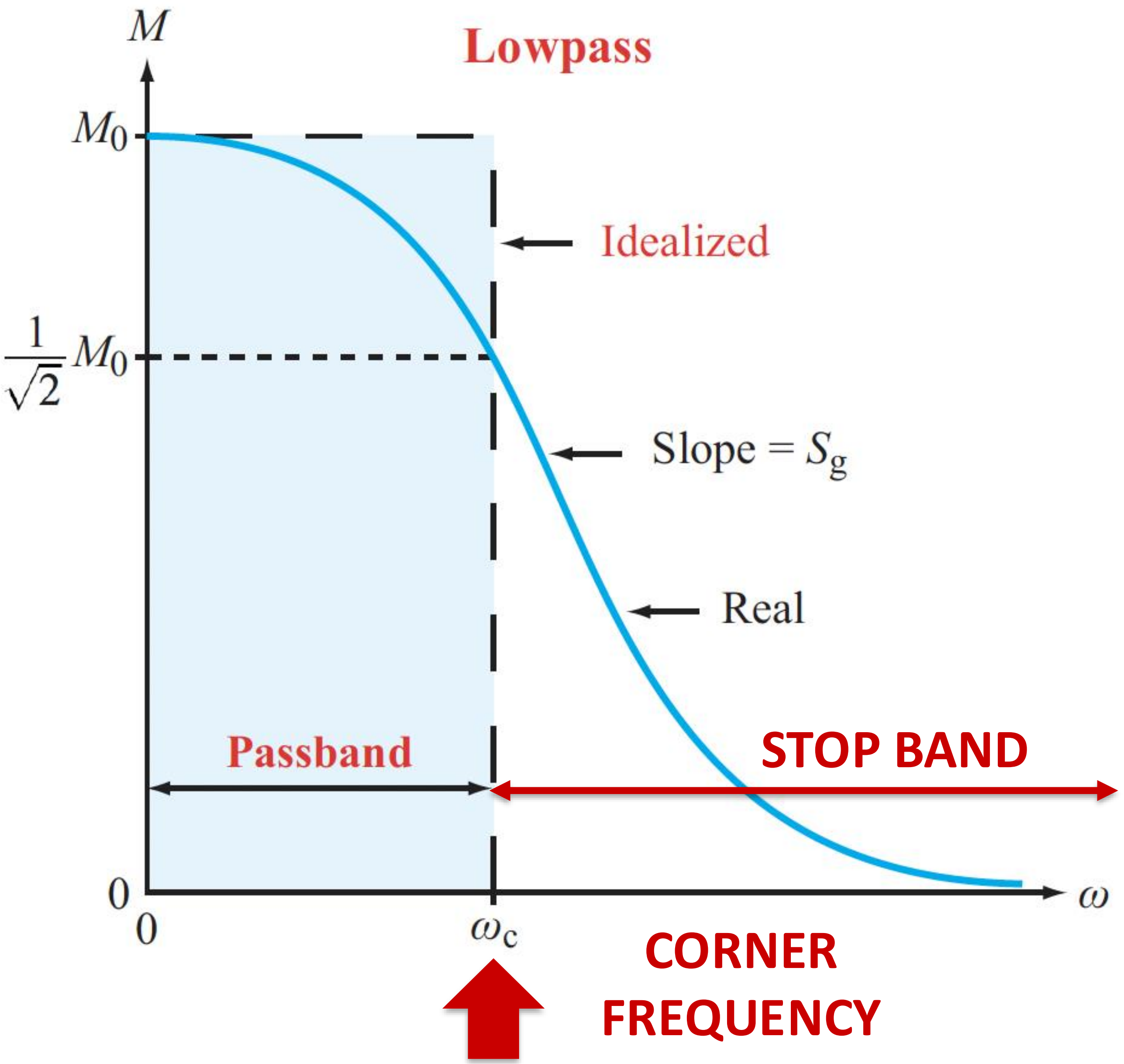




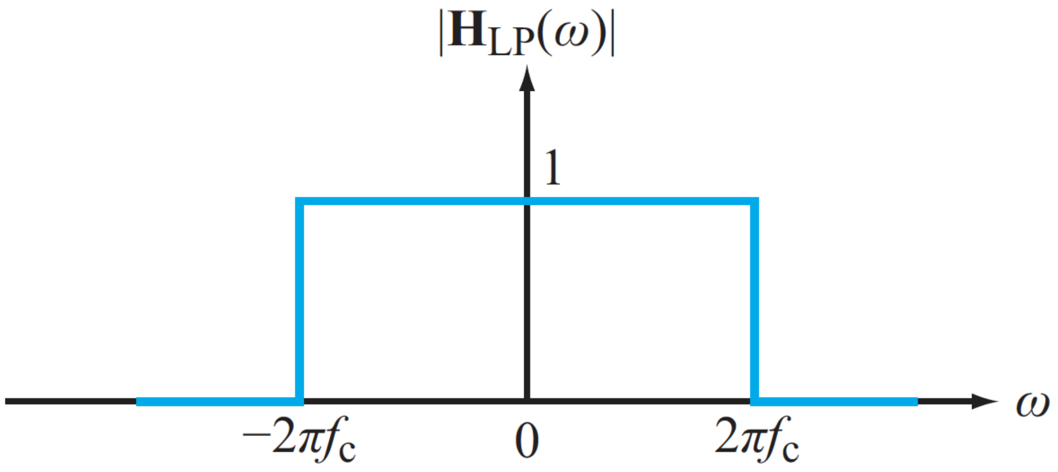


Duality in action

The Low-pass Filter

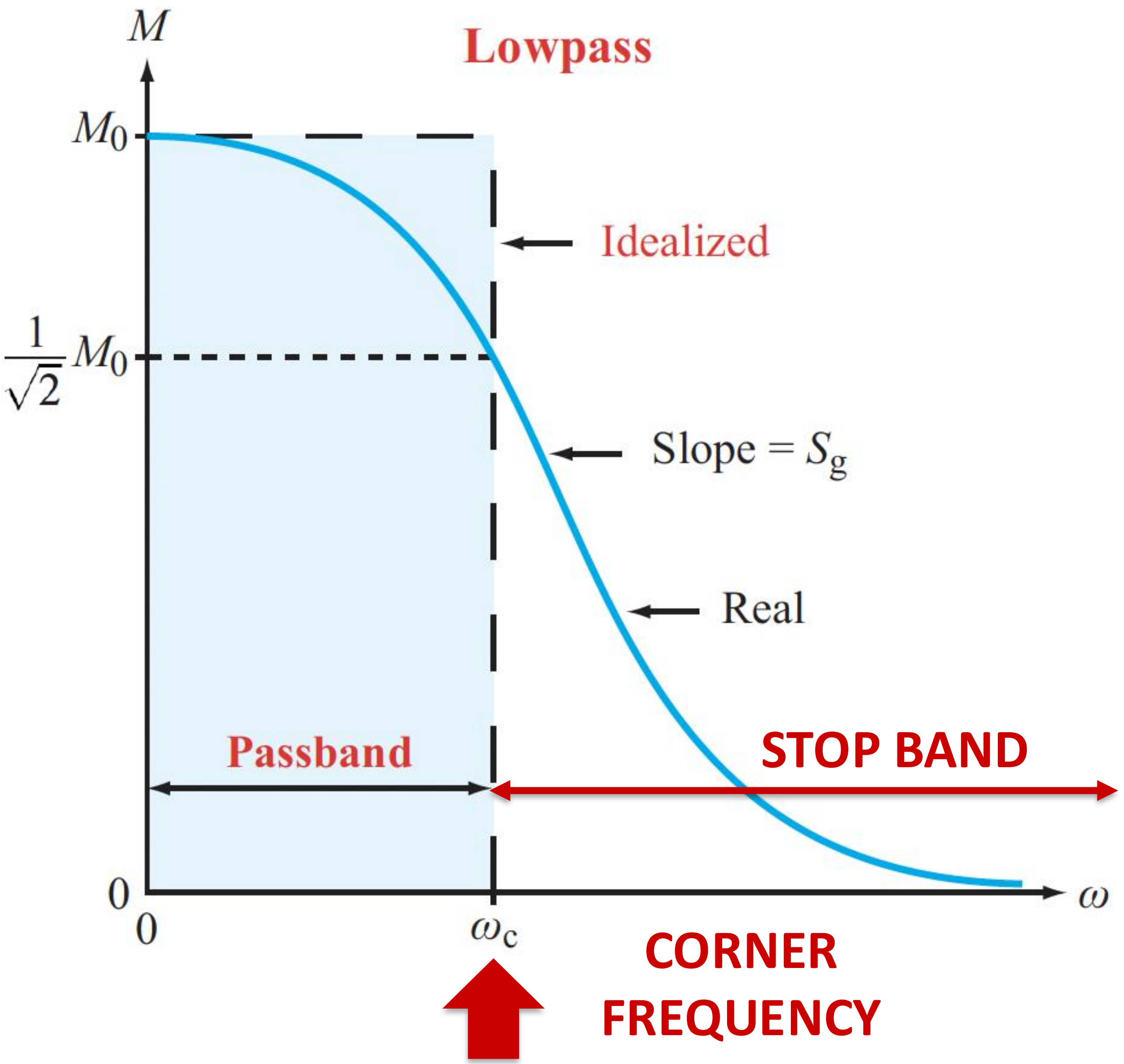


The ideal low-pass filter is a "pulse" in the frequency domain



- Unfortunately, the time-domain inverse (sinc function) is noncausal, so an ideal Low-pass filter is impossible to build
- This is also true for the four other idealized filters

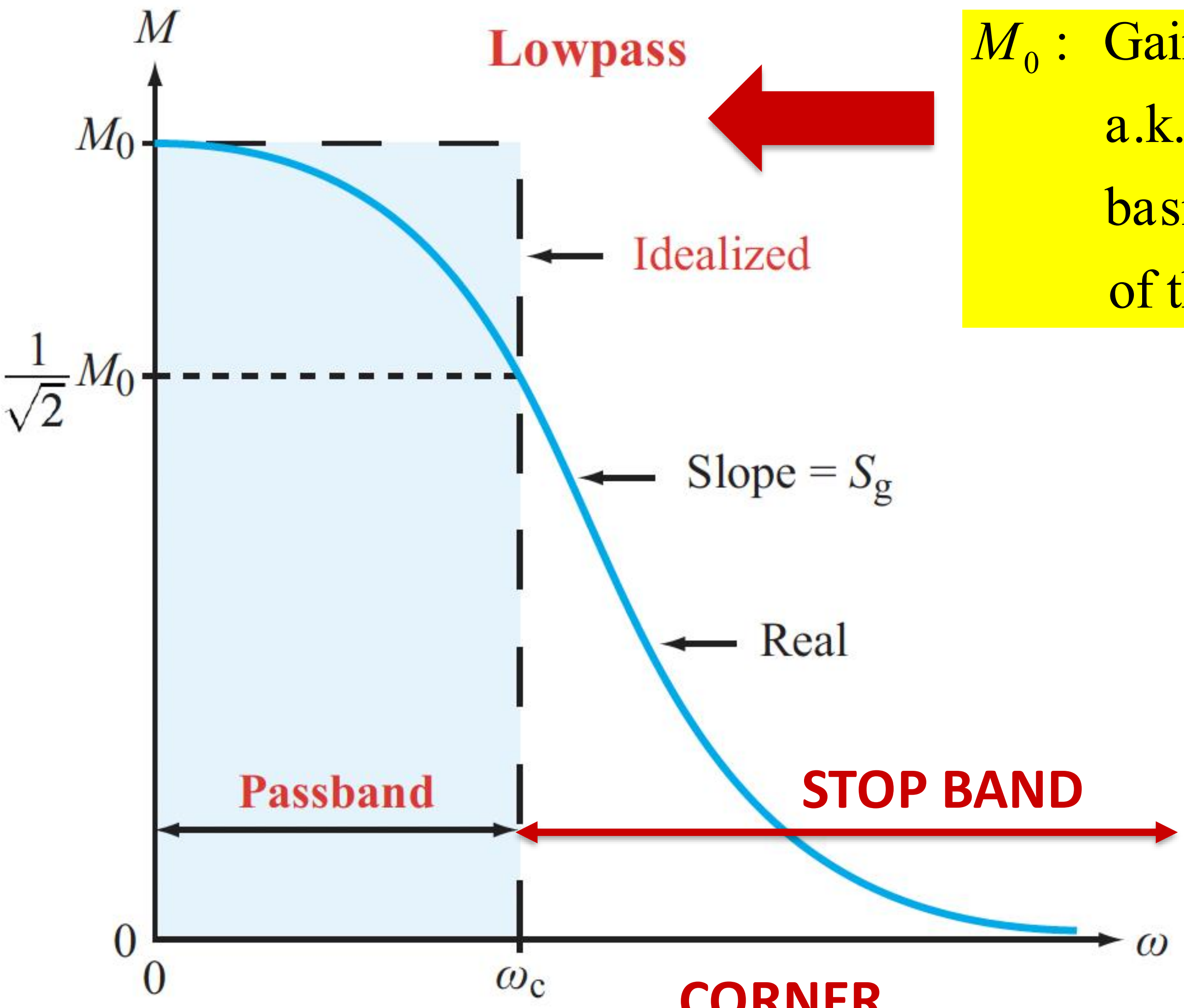
The Low-pass Filter



We can, at best, approximate the ideal filter

- This is the field of "filter design"
- Filter design is all about making tradeoffs to create a filter that meets given specifications
  - Steepness of slope
  - Flatness of passband
  - Flatness of stop band
  - Cost
  - Etc.

The Low-pass Filter



M_0 : Gain factor

a.k.a "dc Gain"

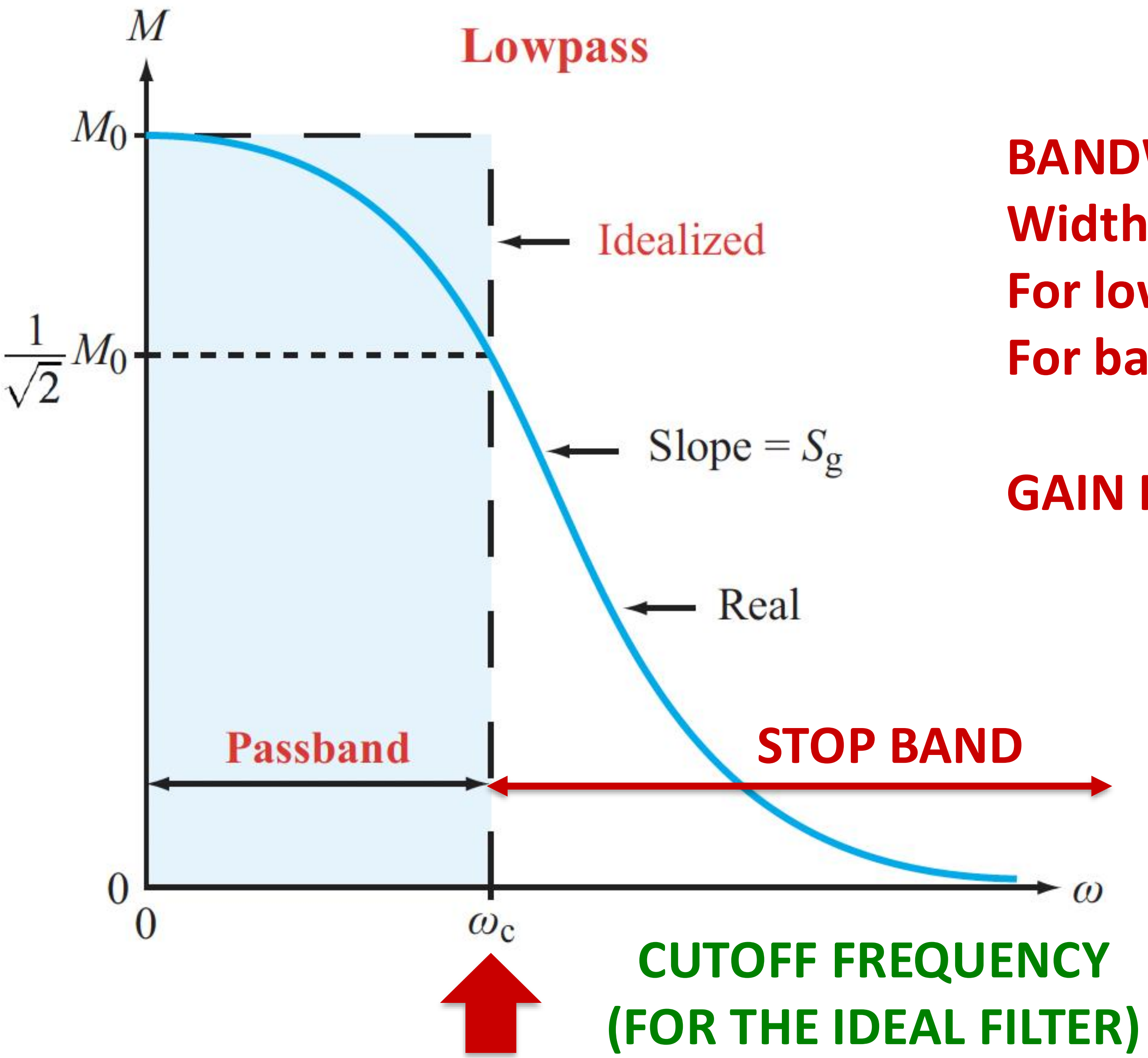
basically, the reference "nominal value" of the passband

$$M^2(\omega_C) = \frac{M_0^2}{2}$$

$$M(\omega_C) = \frac{M_0}{\sqrt{2}}$$

CORNER FREQUENCY ⟷ HALF POWER FREQUENCY

The Low-pass Filter



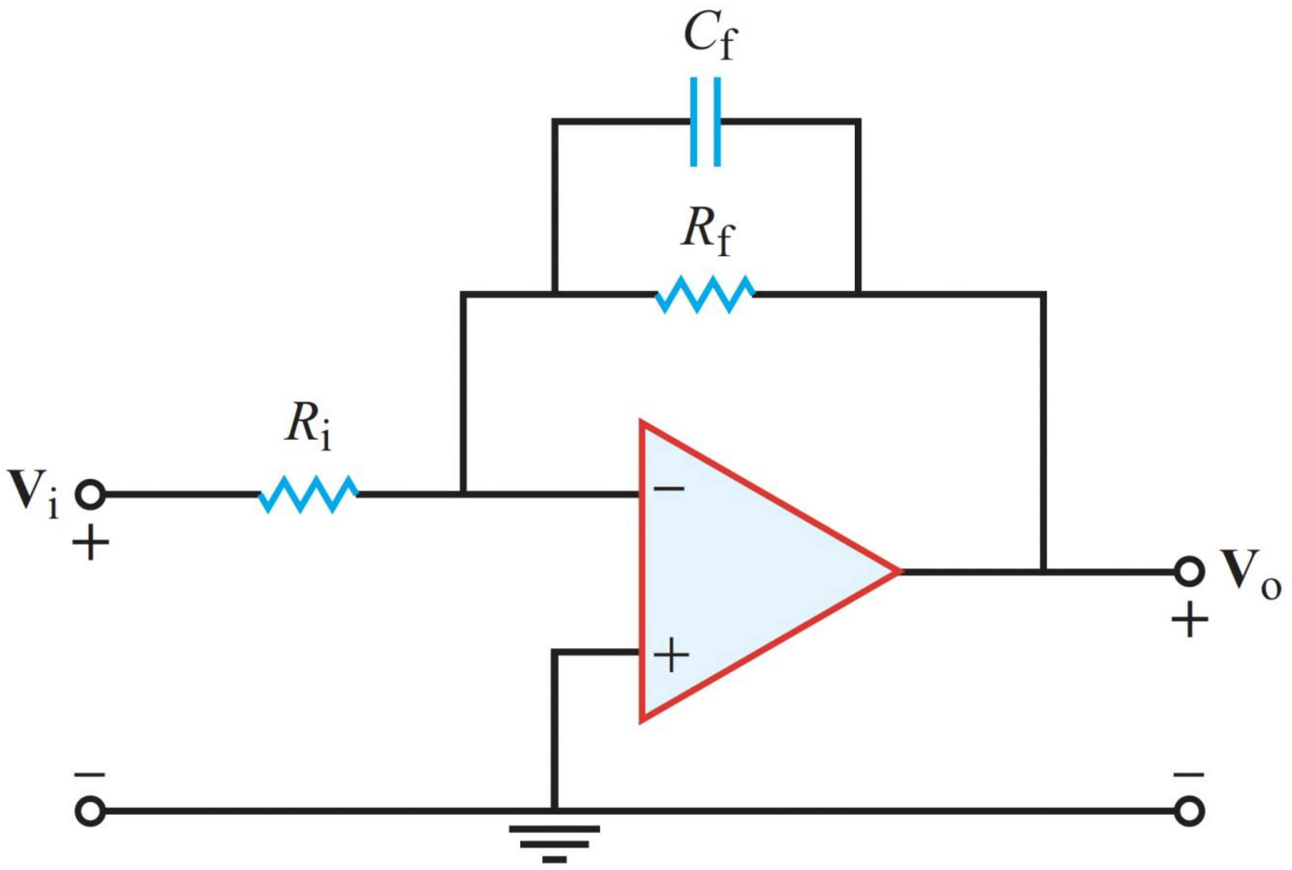
BANDWIDTH:
Width of the filter's "working" band
For low-pass: width of pass band
For band reject: width of the rejection band

GAIN ROLL-OFF RATE = Slope at ω_c = S_g

Intuition:

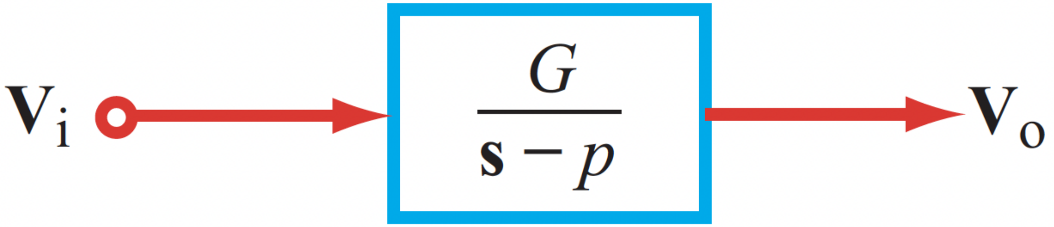
- **Higher bandwidth implies signals that can change faster**
- **Communication channels with higher bandwidth can carry more information**

Activity: Single-pole Low-pass Op-amp Circuit



$$\mathbf{H}(\mathbf{s}) = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i} = -\left(\frac{R_f}{R_i}\right)\left[\frac{1}{1 + R_f C_f \mathbf{s}}\right]$$

System Representation:



$$G = -\frac{1}{R_i C_f}, \quad p = -\frac{1}{R_f C_f}$$

For this filter, derive the expressions for $M(\omega), \varphi(\omega), M_0, \omega_C$

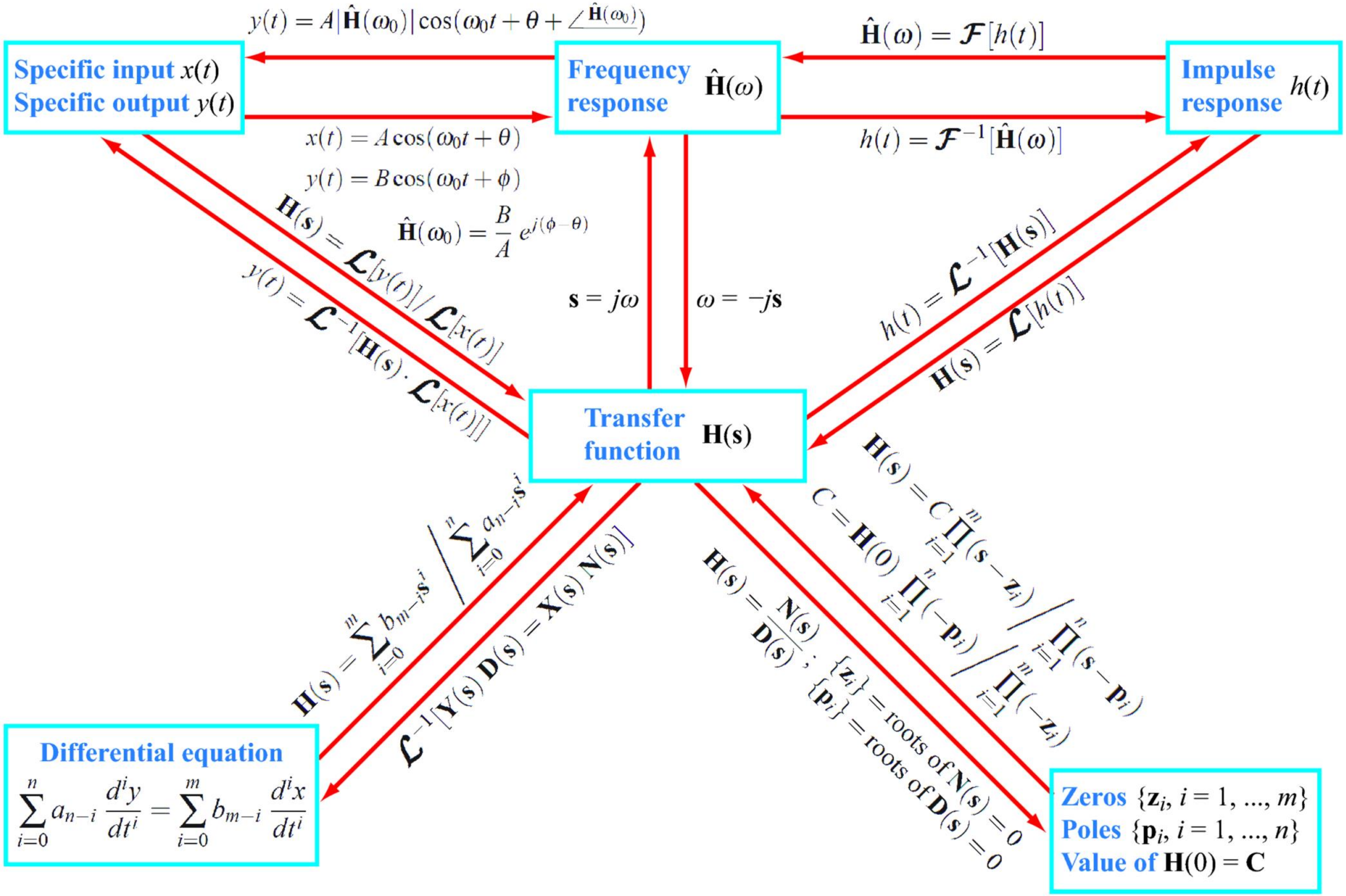
Summary

Today, we learned to use Fourier Transforms for analyzing circuits, and gave LTI systems a new name - filters

- The principles and methods are mostly similar to the use of Laplace transforms
- A good way to handle non-causal signals and systems
- Choose the best approach for a given problem:

Input $x(t)$		**Solution Method**	**Output $y(t)$**
Duration	**Waveform**		
Everlasting	**Sinusoid**	Phasor Domain	Steady State Component (no transient exists)
Everlasting	**Periodic**	Phasor Domain and Fourier Series	Steady State Component (no transient exists)
Causal, $x(t) = 0$, for $t < 0$	**Any**	Laplace Transform (unilateral) (can accommodate non-zero initial conditions)	Complete Solution (transient + steady state)
Everlasting	**Any**	Bilateral Laplace Transform or Fourier Transform	Complete Solution (transient + steady state)

Broader Recap



For the Next Class

- Practice using Tables 5.6 and 5.7.
- Try Exercise 5-9 from your textbook
  - Asks you to calculate the capacitor voltages for the in-class examples
  - The answers are in the textbook itself
- The Canvas site has a bunch of hand-worked Fourier series examples, and complete solutions to Problems 5.26 and 5.60 of your textbook.
  - Use them for practice
- Study ahead for Section 6.1 – 6.2
  - This is a fun chapter!